ACTIVITY – 1

REGISTER NUMBER: 192424166

DATE: 20-07-2026

VISUALIZATION EVALUATION FOR DATA ANALYSIS SCENARIOS

OBJECTIVE: TO EVALUATE DIFFERENT DATA VISUALIZATIONS FOR COMMUNICATING WORKLOAD, VARIABILITY, DISTRIBUTION, TRENDS, AND DECISION-MAKING INSIGHTS ACROSS REAL-WORLD APPLICATION SCENARIOS.

1. HOSPITAL PATIENT ADMISSION ANALYSIS

A HOSPITAL HAS COLLECTED PATIENT ADMISSION DATA ACROSS CARDIOLOGY, NEUROLOGY, PEDIATRICS, ORTHOPEDICS, AND EMERGENCY DEPARTMENTS FOR ONE YEAR. THE MAIN OBJECTIVE IS TO UNDERSTAND DEPARTMENT WORKLOAD, ADMISSION VARIABILITY, AND PATIENT DISTRIBUTION.

EVALUATION OF VISUALIZATIONS

VISUALIZATION	EVALUATION AND EVIDENCE
BAR CHART	BEST FOR DIRECTLY COMPARING TOTAL ADMISSIONS BETWEEN DEPARTMENTS. IT CLEARLY IDENTIFIES DEPARTMENTS WITH THE HIGHEST AND LOWEST WORKLOAD.
STACKED BAR CHART	USEFUL WHEN ADMISSIONS ARE DIVIDED INTO CATEGORIES SUCH AS MONTHS, AGE GROUPS, OR PATIENT TYPES. IT SHOWS BOTH TOTALS AND COMPOSITION, BUT CAN BECOME DIFFICULT TO COMPARE INDIVIDUAL SEGMENTS.
TREEMAP	PROVIDES A COMPACT VIEW OF PROPORTIONAL PATIENT DISTRIBUTION. IT IS VISUALLY ATTRACTIVE FOR SHOWING CONTRIBUTION TO TOTAL ADMISSIONS, BUT EXACT COMPARISONS ARE HARDER.
BOX PLOT	BEST FOR ANALYZING VARIABILITY IN MONTHLY ADMISSIONS AND DETECTING UNUSUALLY HIGH OR LOW ADMISSION PERIODS. IT DOES NOT DIRECTLY SHOW TOTAL WORKLOAD.

Histogram	Shows the frequency distribution of admission counts and helps identify skewness and common ranges, but is less effective for comparing departments directly.

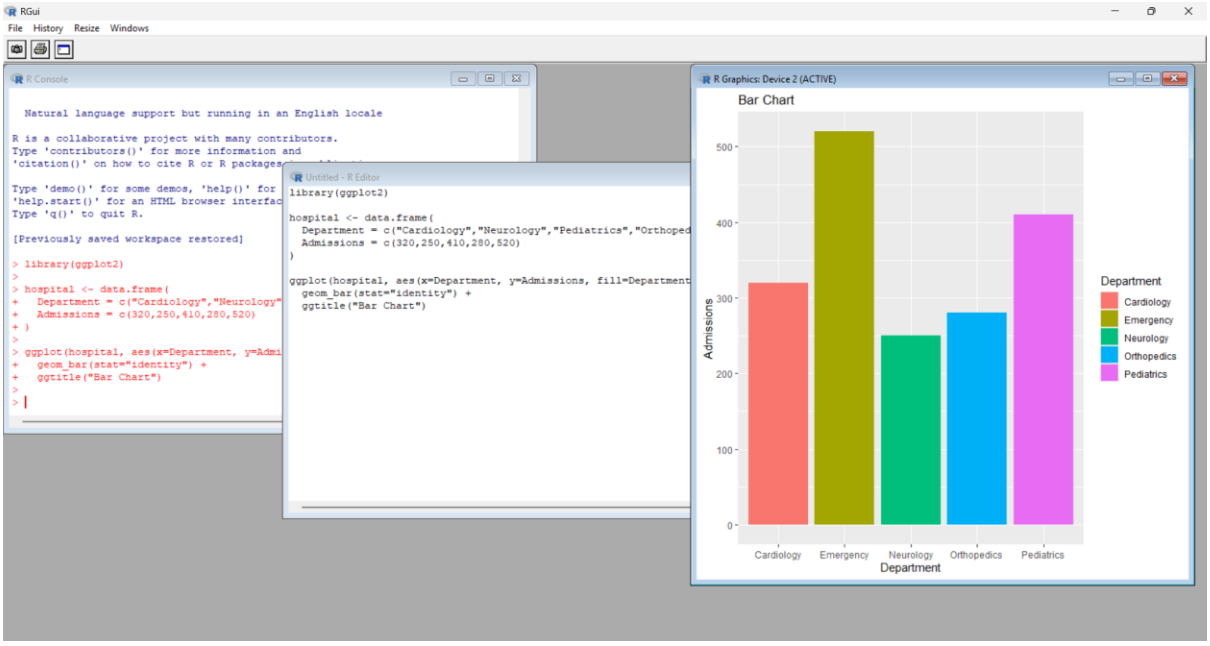
Recommendation: A combination of a **bar chart** and **box plot** is most effective. The bar chart communicates department workload clearly, while the box plot reveals

VARIABILITY AND OUTLIERS. IF ONLY ONE VISUALIZATION MUST BE SELECTED FOR EXECUTIVES, THE BAR CHART IS THE CLEAREST BECAUSE WORKLOAD COMPARISON IS THE PRIMARY REQUIREMENT.

SUGGESTED LIBRARIES: PYTHON: PANDAS, MATPLOTLIB, SEABORN, PLOTLY | R: GGPLOT2, DPLYR

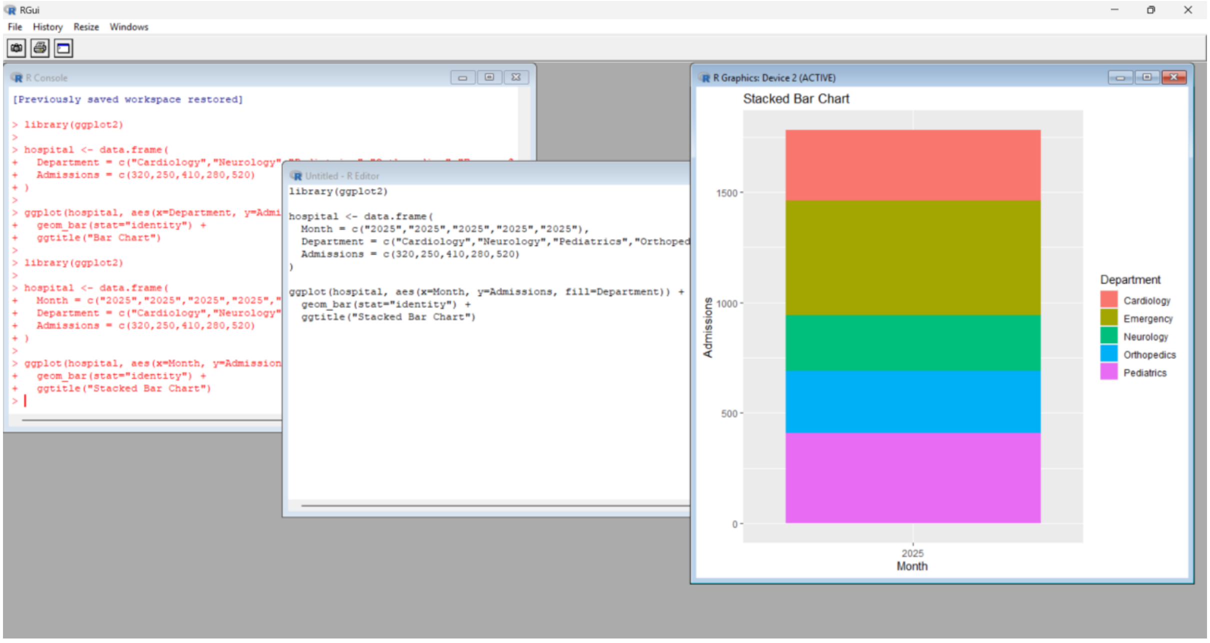
1.1 BAR CHART

```
LIBRARY(GGPLOT2)

HOSPITAL <- DATA.FRAME(
  DEPARTMENT =
C("CARDIOLOGY","NEUROLOGY","PEDIATRICS","ORTHOPEDICS","EMERGENCY"),
  ADMISSIONS = C(320,250,410,280,520)
)

GGPLOT(HOSPITAL, AES(X=DEPARTMENT, Y=ADMISSIONS, FILL=DEPARTMENT)) +
  GEOM_BAR(STAT="IDENTITY") +
  GGTITLE("BAR CHART")
```



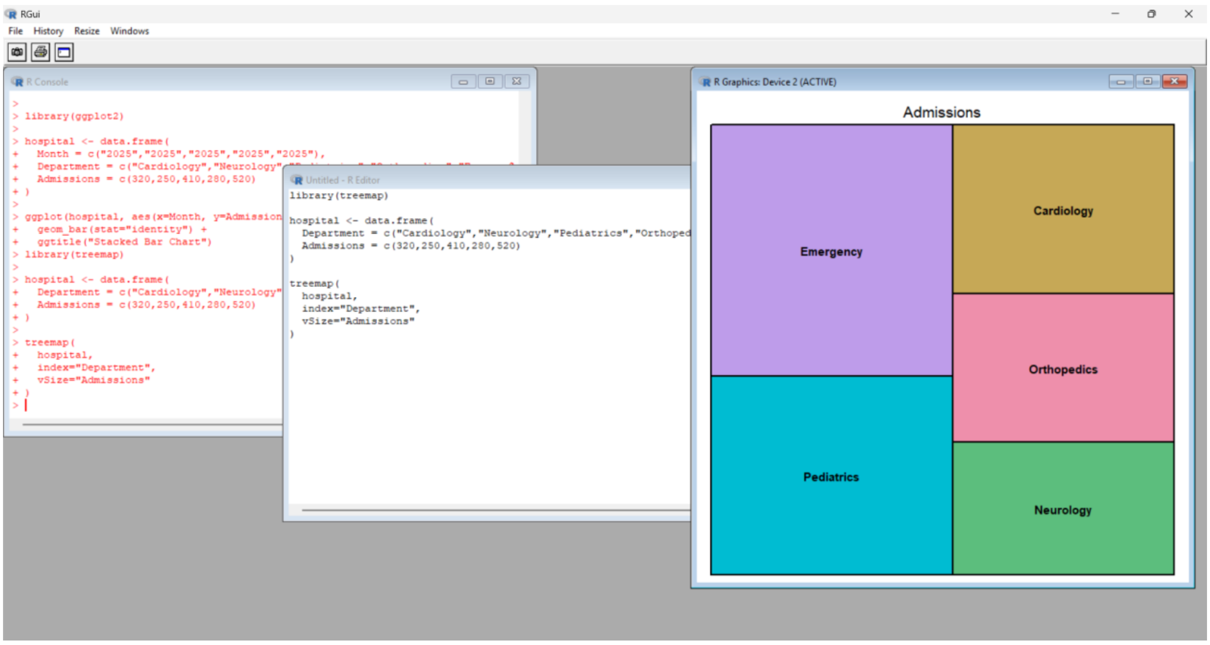
1.2STACKED BAR CHART

```
LIBRARY(GGPLOT2)

HOSPITAL <- DATA.FRAME(
  MONTH = C("2025","2025","2025","2025","2025"),
  DEPARTMENT =
C("CARDIOLOGY","NEUROLOGY","PEDIATRICS","ORTHOPEDICS","EMERGENCY"),
  ADMISSIONS = C(320,250,410,280,520)
)

GGPLOT(HOSPITAL, AES(X=MONTH, Y=ADMISSIONS, FILL=DEPARTMENT)) +
  GEOM_BAR(STAT="IDENTITY") +
  GGTITLE("STACKED BAR CHART")
```



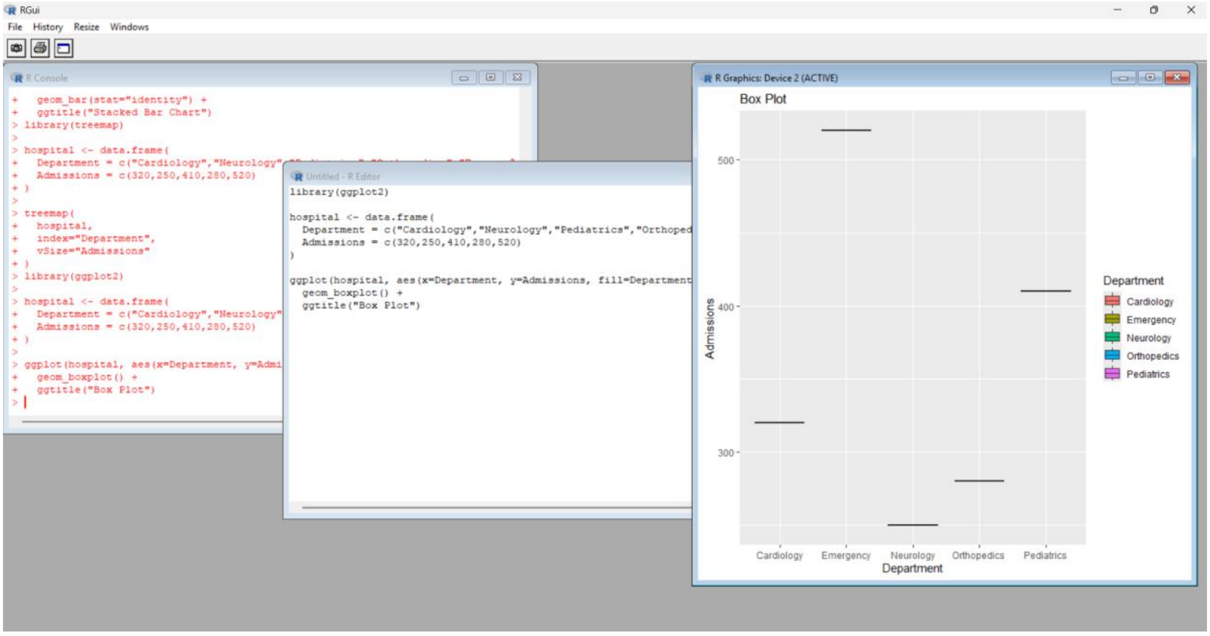
1.3. TREEMAP

```
LIBRARY(TREEMAP)

HOSPITAL <- DATA.FRAME(
  DEPARTMENT =
C("CARDIOLOGY","NEUROLOGY","PEDIATRICS","ORTHOPEDICS","EMERGENCY"),
  ADMISSIONS = C(320,250,410,280,520)
)

TREEMAP(
  HOSPITAL,
  INDEX="DEPARTMENT",
  VSIZE="ADMISSIONS"
)
```



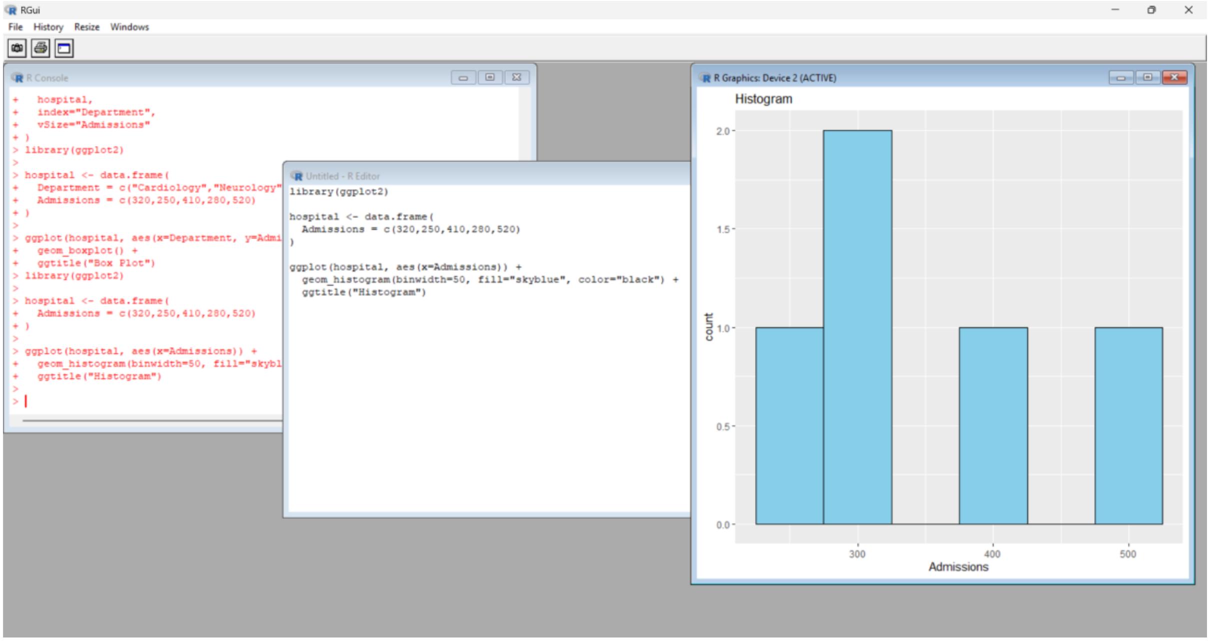
1.4. BOX PLOT

```
LIBRARY(GGPLOT2)

HOSPITAL <- DATA.FRAME(
  DEPARTMENT =
C("CARDIOLOGY","NEUROLOGY","PEDIATRICS","ORTHOPEDICS","EMERGENCY"),
  ADMISSIONS = C(320,250,410,280,520)
)

GGPLOT(HOSPITAL, AES(X=DEPARTMENT, Y=ADMISSIONS, FILL=DEPARTMENT)) +
  GEOM_BOXPLOT() +
  GGTITLE("BOX PLOT")
```



1.5. HISTOGRAM

```
LIBRARY(GGPLOT2)

HOSPITAL <- DATA.FRAME(
  ADMISSIONS = C(320,250,410,280,520)
)

GGPLOT(HOSPITAL, AES(X=ADMISSIONS)) +
  GEOM_HISTOGRAM(BINWIDTH=50, FILL="SKYBLUE", COLOR="BLACK") +
  GGTITLE("HISTOGRAM")
```


RGui
File History Resize Windows
R Console
+ hospital,
+ index="Department",
+ vSize="Admissions"
+)
> library(ggplot2)
>
> hospital <- data.frame(
+ Department = c("Cardiology","Neurology"
+ Admissions = c(320,250,410,280,520)
+)
>
> ggplot(hospital, aes(x=Department, y=Admi
+ geom_boxplot() +
+ ggtitle("Box Plot")
> library(ggplot2)
>
> hospital <- data.frame(
+ Admissions = c(320,250,410,280,520)
+)
>
> ggplot(hospital, aes(x=Admissions)) +
+ geom_histogram(binwidth=50, fill="skybl
+ ggtitle("Histogram")
>
>
Untitled - R Editor
library(ggplot2)
hospital <- data.frame(
Admissions = c(320,250,410,280,520)
)
ggplot(hospital, aes(x=Admissions)) +
geom_histogram(binwidth=50, fill="skyblue", color="black") +
ggtitle("Histogram")
R Graphics: Device 2 (ACTIVE)
Histogram
2.0
1.5
1.0
0.5
0.0
count
300
400
500
Admissions

2. E-COMMERCE CUSTOMER SPENDING BEHAVIOUR

AN ONLINE RETAILER HAS CUSTOMER PURCHASE AMOUNTS, ORDER FREQUENCY, AND PRODUCT CATEGORIES. THE ANALYSIS AIMS TO IDENTIFY HIGH-VALUE CUSTOMERS, UNDERSTAND SPENDING VARIABILITY, AND OBSERVE PURCHASING TRENDS.

EVALUATION OF VISUALIZATIONS

VISUALIZATION	EVALUATION AND EVIDENCE
HISTOGRAM	SHOWS THE DISTRIBUTION OF PURCHASE AMOUNTS AND HELPS IDENTIFY WHETHER SPENDING IS CONCENTRATED AMONG LOW-VALUE CUSTOMERS OR SPREAD ACROSS DIFFERENT RANGES.
VIOLIN PLOT	COMBINES DISTRIBUTION SHAPE WITH SUMMARY STATISTICS AND IS USEFUL FOR COMPARING SPENDING BEHAVIOUR ACROSS PRODUCT CATEGORIES OR CUSTOMER GROUPS.
BOX PLOT	CLEARLY IDENTIFIES MEDIAN SPENDING, VARIABILITY, AND HIGH-VALUE OUTLIERS. IT IS PARTICULARLY USEFUL FOR SPOTTING CUSTOMERS WHOSE SPENDING IS MUCH HIGHER THAN THE TYPICAL CUSTOMER.
DENSITY PLOT	PROVIDES A SMOOTH REPRESENTATION OF SPENDING DISTRIBUTIONS AND MAKES IT EASIER TO COMPARE OVERLAPPING CUSTOMER SEGMENTS.
BAR CHART	BEST FOR COMPARING TOTAL OR AVERAGE SPENDING ACROSS PRODUCT CATEGORIES AND SHOWING PURCHASING TRENDS AT AN AGGREGATED LEVEL.

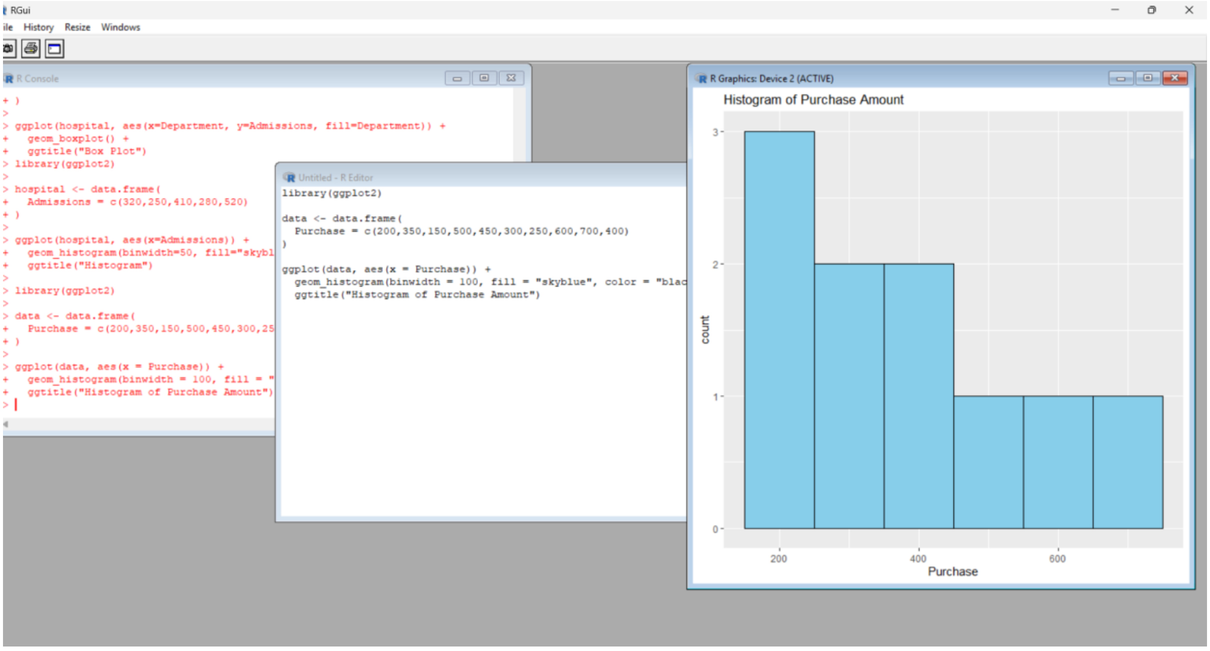
RECOMMENDATION: FOR BUSINESS EXECUTIVES, A **BOX PLOT** IS HIGHLY SUITABLE FOR IDENTIFYING HIGH-VALUE CUSTOMERS AND SPENDING VARIABILITY, WHILE A **BAR CHART** SHOULD COMPLEMENT IT TO COMMUNICATE CATEGORY-LEVEL PURCHASING TRENDS. A DASHBOARD COMBINING BOTH PROVIDES THE STRONGEST BUSINESS INSIGHT.

SUGGESTED LIBRARIES: PYTHON: SEABORN, PLOTLY EXPRESS | R: GGPLOT2

2.1.HISTOGRAM

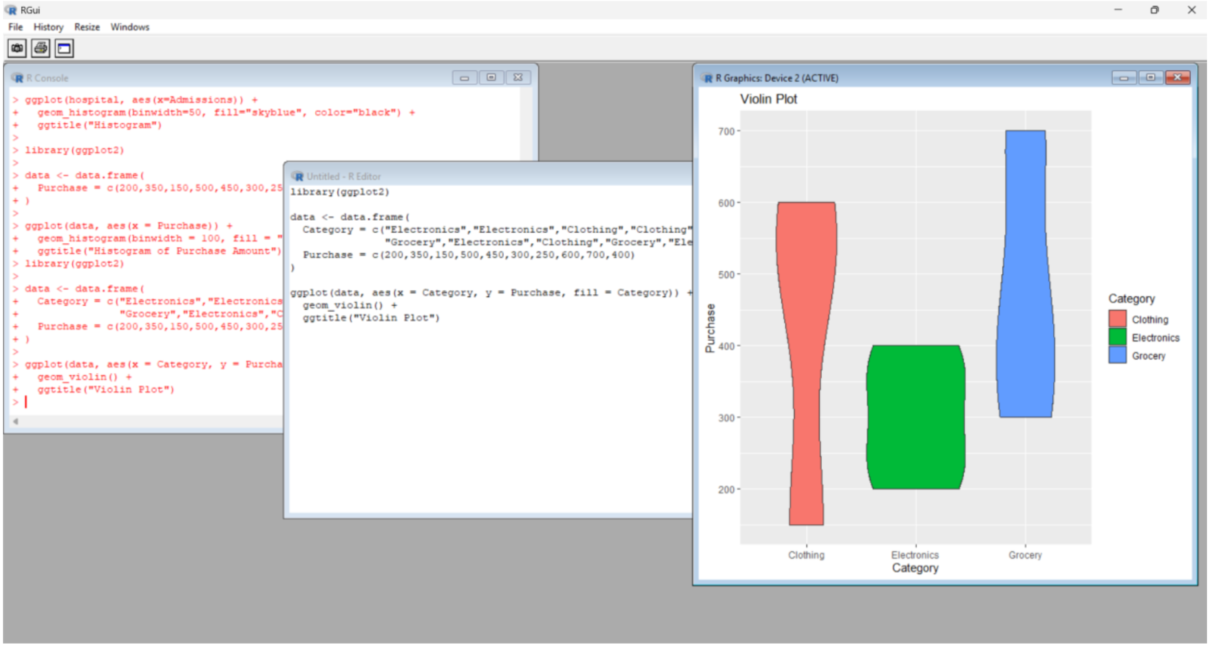
LIBRARY(GGPLOT2)

```
DATA <- DATA.FRAME(
  PURCHASE = C(200,350,150,500,450,300,250,600,700,400)
)

GGPLOT(DATA, AES(X = PURCHASE)) +
  GEOM_HISTOGRAM(BINWIDTH = 100, FILL = "SKYBLUE", COLOR = "BLACK") +
  GGTITLE("HISTOGRAM OF PURCHASE AMOUNT")
```



2.2.VIOLIN PLOT

```
LIBRARY(GGPLOT2)

DATA <- DATA.FRAME(
  CATEGORY =
C("ELECTRONICS","ELECTRONICS","CLOTHING","CLOTHING","GROCERY",
              "GROCERY","ELECTRONICS","CLOTHING","GROCERY","ELECTRONICS"),
  PURCHASE = C(200,350,150,500,450,300,250,600,700,400)
)

GGPLOT(DATA, AES(X = CATEGORY, Y = PURCHASE, FILL = CATEGORY)) +
  GEOM_VIOLIN() +
  GGTITLE("VIOLIN PLOT")
```



2.3.BOX PLOT

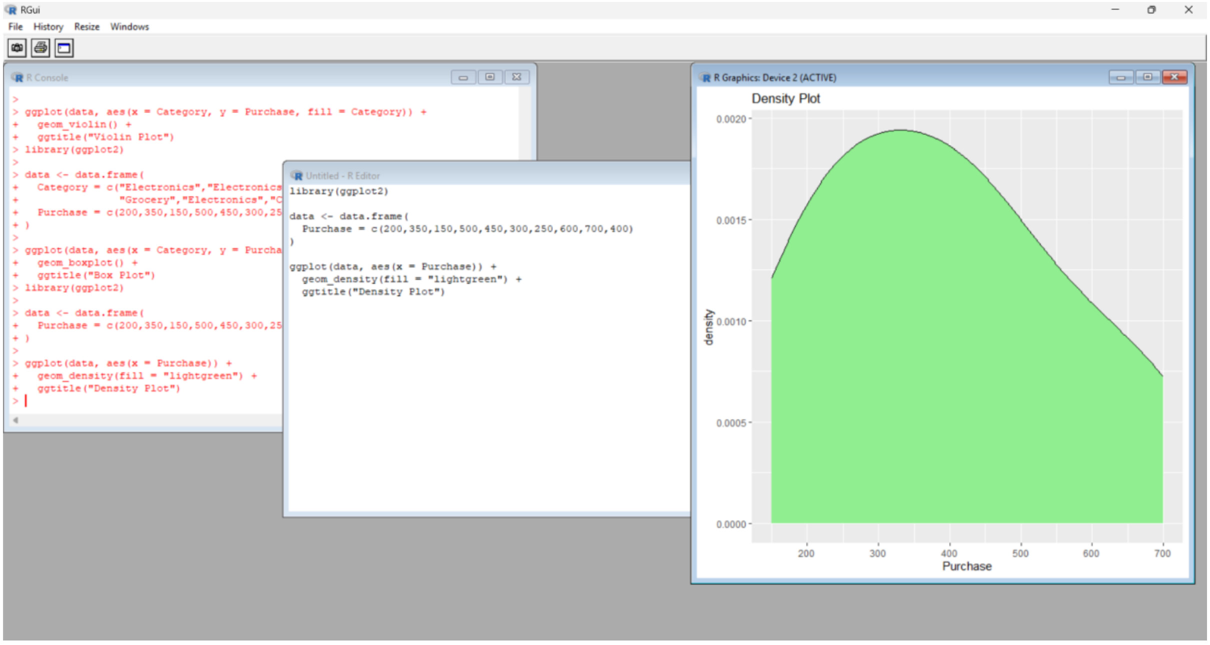
LIBRARY(GGPLOT2)

DATA <- DATA.FRAME(
  CATEGORY =
C("ELECTRONICS","ELECTRONICS","CLOTHING","CLOTHING","GROCERY",
              "GROCERY","ELECTRONICS","CLOTHING","GROCERY","ELECTRONICS"),
  PURCHASE = C(200,350,150,500,450,300,250,600,700,400)
)

GGPLOT(DATA, AES(X = CATEGORY, Y = PURCHASE, FILL = CATEGORY)) +
  GEOM_BOXPLOT() +
  GGTITLE("BOX PLOT")

2.4.DENSITY PLOT

```
LIBRARY(GGPLOT2)

DATA <- DATA.FRAME(
  PURCHASE = C(200,350,150,500,450,300,250,600,700,400)
)

GGPLOT(DATA, AES(X = PURCHASE)) +
  GEOM_DENSITY(FILL = "LIGHTGREEN") +
  GGTITLE("DENSITY PLOT")
```



2.5.BAR CHART

```
LIBRARY(GGPLOT2)

DATA <- DATA.FRAME(
  CATEGORY = C("ELECTRONICS","CLOTHING","GROCERY"),
  ORDERS = C(40,30,50)
)

GGPLOT(DATA, AES(X = CATEGORY, Y = ORDERS, FILL = CATEGORY)) +
  GEOM_BAR(STAT = "IDENTITY") +
  GGTITLE("BAR CHART")
```


RGui
File History Resize Windows
R Console
+)
>
> ggplot(data, aes(x = Category, y = Purchase, fill = Category)) +
+ geom_boxplot() +
+ ggtitle("Box Plot")
> library(ggplot2)
>
> data <- data.frame(
+ Purchase = c(200,350,150,500,450,300,25
+)
>
> ggplot(data, aes(x = Purchase)) +
+ geom_density(fill = "lightgreen") +
+ ggtitle("Density Plot")
> library(ggplot2)
>
> data <- data.frame(
+ Category = c("Electronics","Clothing","
+ Orders = c(40,30,50)
+)
>
> ggplot(data, aes(x = Category, y = Orders
+ geom_bar(stat = "identity") +
+ ggtitle("Bar Chart")
>
Untitled - R Editor
library(ggplot2)
data <- data.frame(
Category = c("Electronics","Clothing","Grocery"),
Orders = c(40,30,50)
)
ggplot(data, aes(x = Category, y = Orders, fill = Category)) +
geom_bar(stat = "identity") +
ggtitle("Bar Chart")
R Graphics: Device 2 (ACTIVE)
Bar Chart
Orders
0
10
20
30
40
50
Clothing
Electronics
Grocery
Category
Category
Clothing
Electronics
Grocery

3. UNIVERSITY EXAMINATION PERFORMANCE DASHBOARD

A UNIVERSITY WANTS TO COMPARE STUDENT MARKS ACROSS DEPARTMENTS, IDENTIFY SCORE DISTRIBUTIONS, DETECT OUTLIERS, AND RECOGNIZE TOP-PERFORMING CLASSES. THE DASHBOARD SHOULD SUPPORT ACADEMIC DECISION-MAKING.

EVALUATION OF VISUALIZATIONS

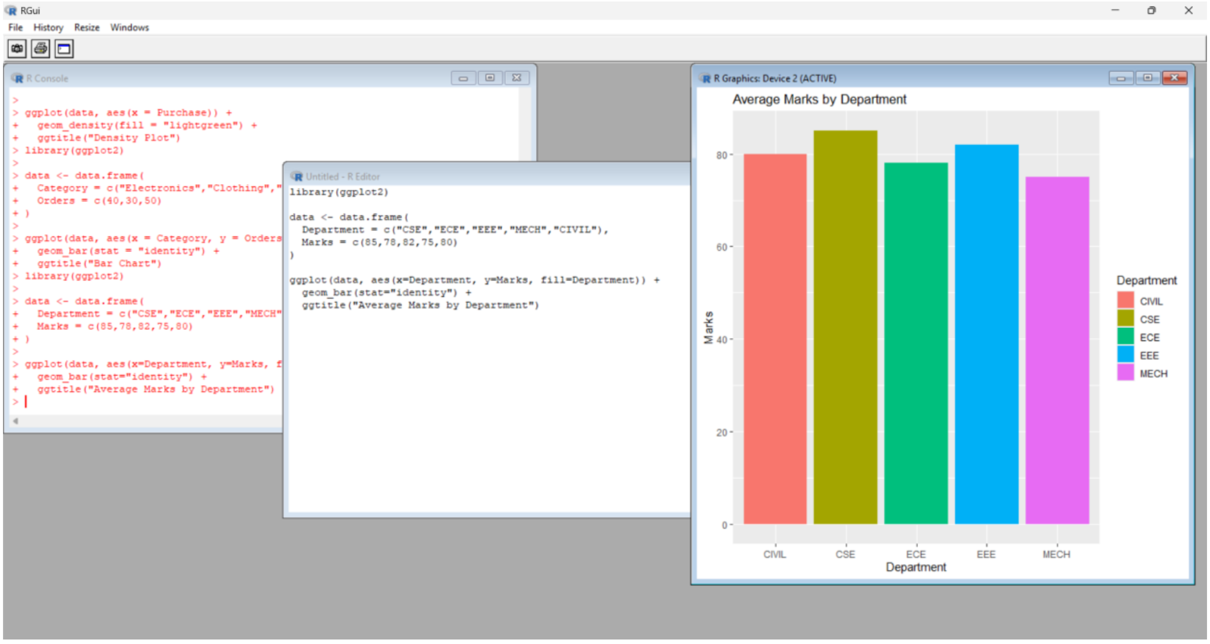
VISUALIZATION	EVALUATION AND EVIDENCE
BAR CHART	CLEARLY COMPARES AVERAGE OR MEDIAN MARKS ACROSS DEPARTMENTS AND CLASSES. IT IS EASY FOR ADMINISTRATORS AND ACADEMIC STAFF TO INTERPRET.
BOX PLOT	SHOWS MEDIAN, SCORE SPREAD, AND OUTLIERS FOR EACH DEPARTMENT. IT IS ONE OF THE MOST EFFECTIVE CHARTS FOR COMPARING EXAMINATION PERFORMANCE DISTRIBUTIONS.
HISTOGRAM	SHOWS THE OVERALL FREQUENCY DISTRIBUTION OF MARKS AND HELPS IDENTIFY WHETHER SCORES ARE CONCENTRATED, NORMALLY DISTRIBUTED, OR SKEWED.
VIOLIN PLOT	PROVIDES DETAILED DISTRIBUTION SHAPES AND ALLOWS COMPARISON OF SCORE DENSITY ACROSS DEPARTMENTS, ALTHOUGH IT MAY BE LESS FAMILIAR TO GENERAL USERS.
SCATTER PLOT	CAN REVEAL RELATIONSHIPS BETWEEN MARKS AND VARIABLES SUCH AS ATTENDANCE OR ASSIGNMENT SCORES WHEN THOSE VARIABLES ARE AVAILABLE.

RECOMMENDATION: THE DASHBOARD SHOULD PRIMARILY USE **BOX PLOTS** FOR DEPARTMENTAL PERFORMANCE COMPARISON AND OUTLIER DETECTION, SUPPORTED BY **BAR CHARTS** FOR AVERAGE SCORES AND **HISTOGRAMS** FOR OVERALL SCORE DISTRIBUTION. DASHBOARD IMPROVEMENTS SHOULD INCLUDE FILTERS FOR DEPARTMENT, SEMESTER, AND CLASS; CONSISTENT SCALES; CLEAR LABELS; AND KPI CARDS FOR AVERAGE SCORE, PASS PERCENTAGE, AND TOP-PERFORMING CLASS.

SUGGESTED LIBRARIES: PYTHON: PANDAS, SEABORN, PLOTLY | R: GGPLOT2, PLOTLY

3.1. BAR CHART

```
LIBRARY(GGPLOT2)

DATA <- DATA.FRAME(
  DEPARTMENT = C("CSE","ECE","EEE","MECH","CIVIL"),
  MARKS = C(85,78,82,75,80)
)

GGPLOT(DATA, AES(X=DEPARTMENT, Y=MARKS, FILL=DEPARTMENT)) +
  GEOM_BAR(STAT="IDENTITY") +
  GGTITLE("AVERAGE MARKS BY DEPARTMENT")
```



3.2. BOX PLOT

```
LIBRARY(GGPLOT2)

DATA <- DATA.FRAME(
  DEPARTMENT = C("CSE","CSE","ECE","ECE","EEE","EEE","MECH","MECH","CIVIL","CIVIL"),
  MARKS = C(85,90,78,82,80,84,72,78,76,83)
)

GGPLOT(DATA, AES(X=DEPARTMENT, Y=MARKS, FILL=DEPARTMENT)) +
  GEOM_BOXPLOT() +
  GGTITLE("MARKS DISTRIBUTION")
```

3.3. HISTOGRAM

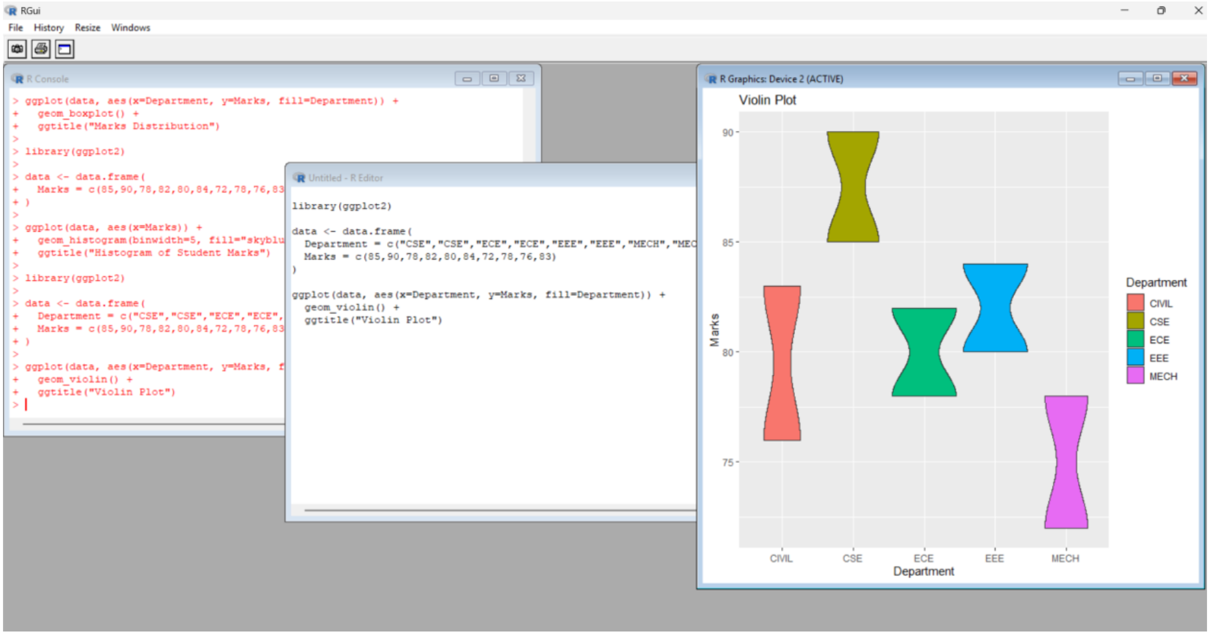
LIBRARY(GGPLOT2)

DATA <- DATA.FRAME(
  MARKS = C(85,90,78,82,80,84,72,78,76,83)
)

GGPLOT(DATA, AES(X=MARKS)) +
  GEOM_HISTOGRAM(BINWIDTH=5, FILL="SKYBLUE", COLOR="BLACK") +
  GGTITLE("HISTOGRAM OF STUDENT MARKS")

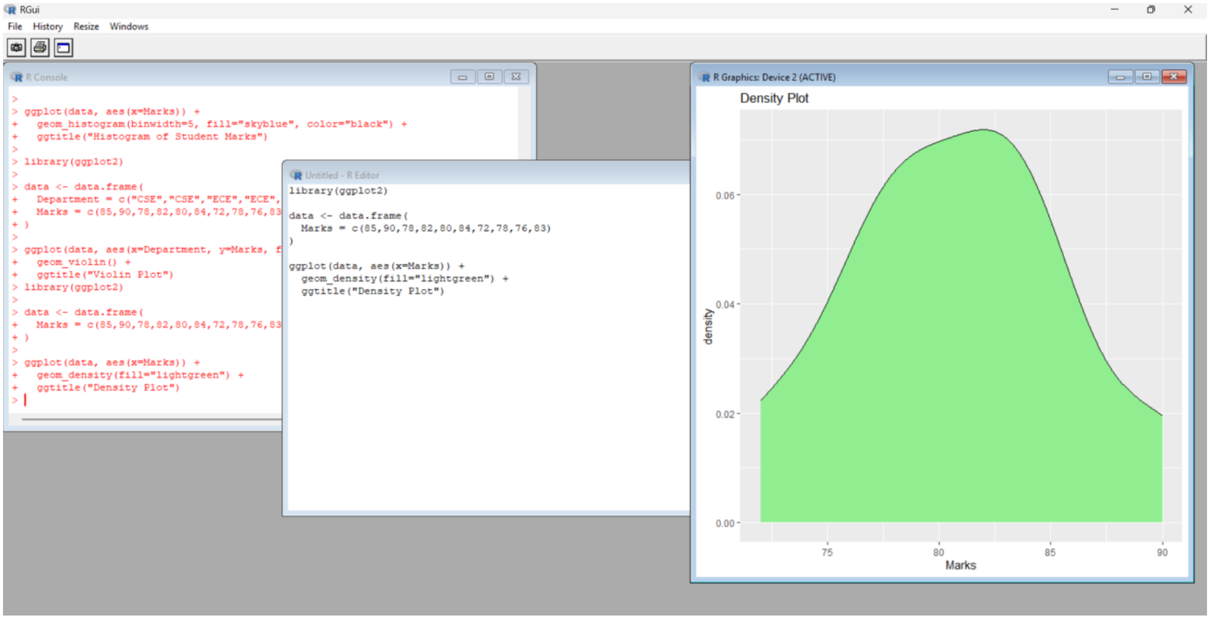
3.4. VIOLIN PLOT

```
LIBRARY(GGPLOT2)

DATA <- DATA.FRAME(
  DEPARTMENT = C("CSE","CSE","ECE","ECE","EEE","EEE","MECH","MECH","CIVIL","CIVIL"),
  MARKS = C(85,90,78,82,80,84,72,78,76,83)
)

GGPLOT(DATA, AES(X=DEPARTMENT, Y=MARKS, FILL=DEPARTMENT)) +
  GEOM_VIOLIN() +
  GGTITLE("VIOLIN PLOT")
```



3.5. DENSITY PLOT

```
LIBRARY(GGPLOT2)

DATA <- DATA.FRAME(
  MARKS = C(85,90,78,82,80,84,72,78,76,83)
)

GGPLOT(DATA, AES(X=MARKS)) +
  GEOM_DENSITY(FILL="LIGHTGREEN") +
  GGTITLE("DENSITY PLOT")
```


RGui
File History Resize Windows
R Console
> ggplot(data, aes(x=Marks)) +
+ geom_histogram(binwidth=5, fill="skyblue", color="black") +
+ ggtitle("Histogram of Student Marks")
> library(ggplot2)
> data <- data.frame(
+ Department = c("CSE","CSE","ECE","ECE",
+ Marks = c(85,90,78,82,80,84,72,78,76,83
+)
> ggplot(data, aes(x=Department, y=Marks, f
+ geom_violin() +
+ ggtitle("Violin Plot")
> library(ggplot2)
> data <- data.frame(
+ Marks = c(85,90,78,82,80,84,72,78,76,83
+)
> ggplot(data, aes(x=Marks)) +
+ geom_density(fill="lightgreen") +
+ ggtitle("Density Plot")
Untitled - R Editor
library(ggplot2)
data <- data.frame(
Marks = c(85,90,78,82,80,84,72,78,76,83)
)
ggplot(data, aes(x=Marks)) +
geom_density(fill="lightgreen") +
ggtitle("Density Plot")
R Graphics: Device 2 (ACTIVE)
Density Plot
density
0.06
0.04
0.02
0.00
75
80
85
90
Marks

4. CLIMATE AND RAINFALL DISTRIBUTION STUDY

METEOROLOGICAL DATA CONTAINS MONTHLY RAINFALL, TEMPERATURE, AND HUMIDITY FOR MULTIPLE CITIES. THE OBJECTIVE IS TO REPRESENT SEASONAL RAINFALL VARIATION AND CLIMATE DISTRIBUTIONS FOR ENVIRONMENTAL RESEARCH.

EVALUATION OF VISUALIZATIONS

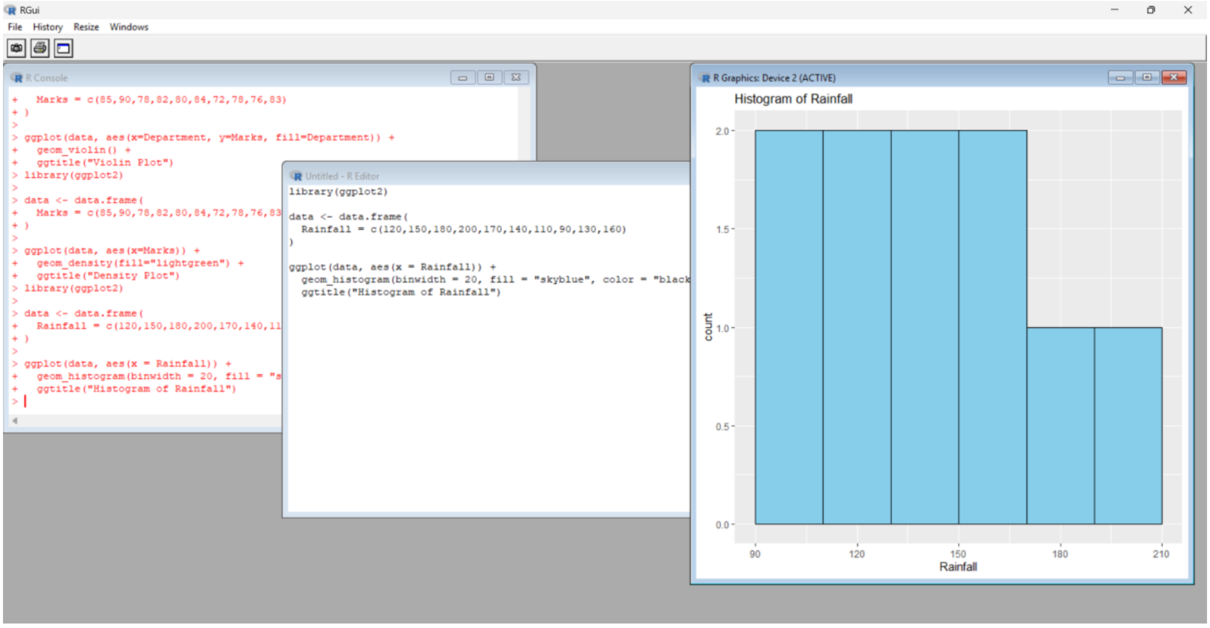
VISUALIZATION	EVALUATION AND EVIDENCE
HISTOGRAM	SHOWS THE FREQUENCY DISTRIBUTION OF RAINFALL OR TEMPERATURE VALUES AND HELPS IDENTIFY COMMON RANGES AND SKEWNESS.
DENSITY PLOT	PROVIDES A SMOOTH VIEW OF THE DISTRIBUTION AND IS USEFUL FOR COMPARING CLIMATE PATTERNS BETWEEN CITIES OR SEASONS.
BOX PLOT	EFFECTIVELY COMPARES MONTHLY OR SEASONAL RAINFALL VARIABILITY, IDENTIFIES MEDIANS, SPREAD, AND EXTREME RAINFALL EVENTS.
BAR CHART	CLEARLY DISPLAYS AVERAGE MONTHLY RAINFALL AND MAKES SEASONAL PATTERNS EASY TO COMMUNICATE, ALTHOUGH IT DOES NOT SHOW DISTRIBUTION OR OUTLIERS AS WELL AS A BOX PLOT.

RECOMMENDATION: FOR ENVIRONMENTAL RESEARCHERS, THE **BOX PLOT** IS THE MOST SUITABLE SINGLE VISUALIZATION FOR COMPARING SEASONAL RAINFALL VARIATION BECAUSE IT DISPLAYS CENTRAL TENDENCY, VARIABILITY, AND EXTREME VALUES. A **BAR CHART** SHOULD BE ADDED WHEN THE PRIMARY GOAL IS TO COMMUNICATE AVERAGE MONTHLY RAINFALL TRENDS. DENSITY PLOTS ARE VALUABLE FOR DETAILED DISTRIBUTION COMPARISONS BETWEEN CITIES.

SUGGESTED LIBRARIES: PYTHON: MATPLOTLIB, SEABORN | R: GGPLOT2

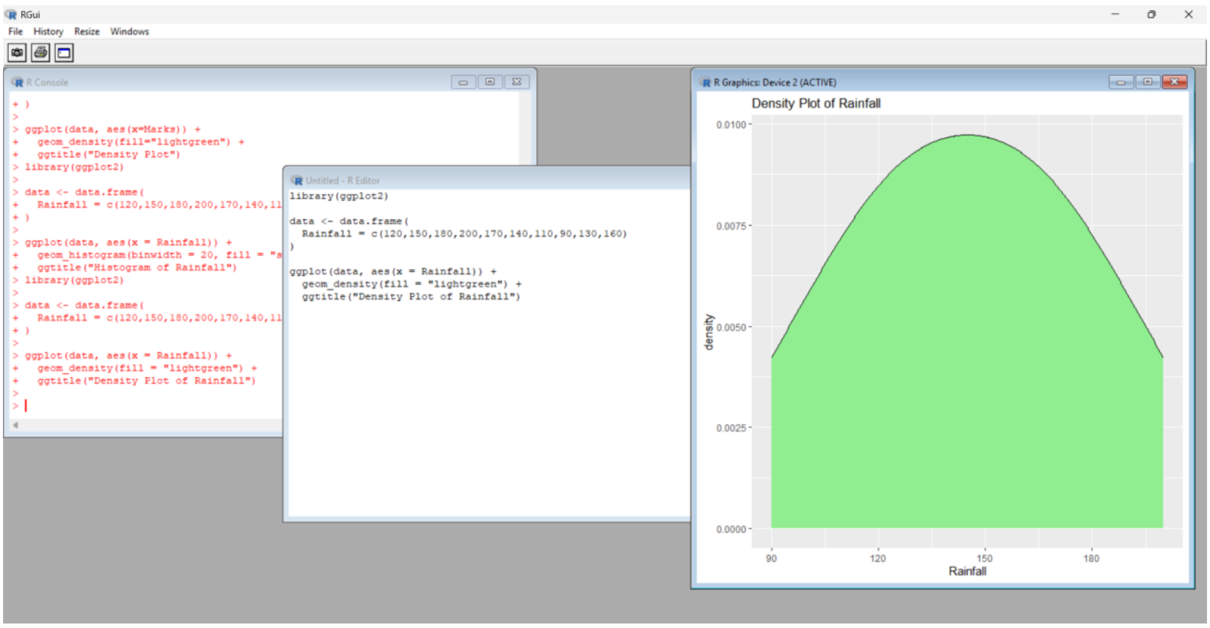
4.1. HISTOGRAM

```
LIBRARY(GGPLOT2)

DATA <- DATA.FRAME(
  RAINFALL = C(120,150,180,200,170,140,110,90,130,160)
)

GGPLOT(DATA, AES(X = RAINFALL)) +
  GEOM_HISTOGRAM(BINWIDTH = 20, FILL = "SKYBLUE", COLOR = "BLACK") +
  GGTITLE("HISTOGRAM OF RAINFALL")
```



4.2. DENSITY PLOT

```
LIBRARY(GGPLOT2)

DATA <- DATA.FRAME(
  RAINFALL = C(120,150,180,200,170,140,110,90,130,160)
)

GGPLOT(DATA, AES(X = RAINFALL)) +
  GEOM_DENSITY(FILL = "LIGHTGREEN") +
  GGTITLE("DENSITY PLOT OF RAINFALL")
```

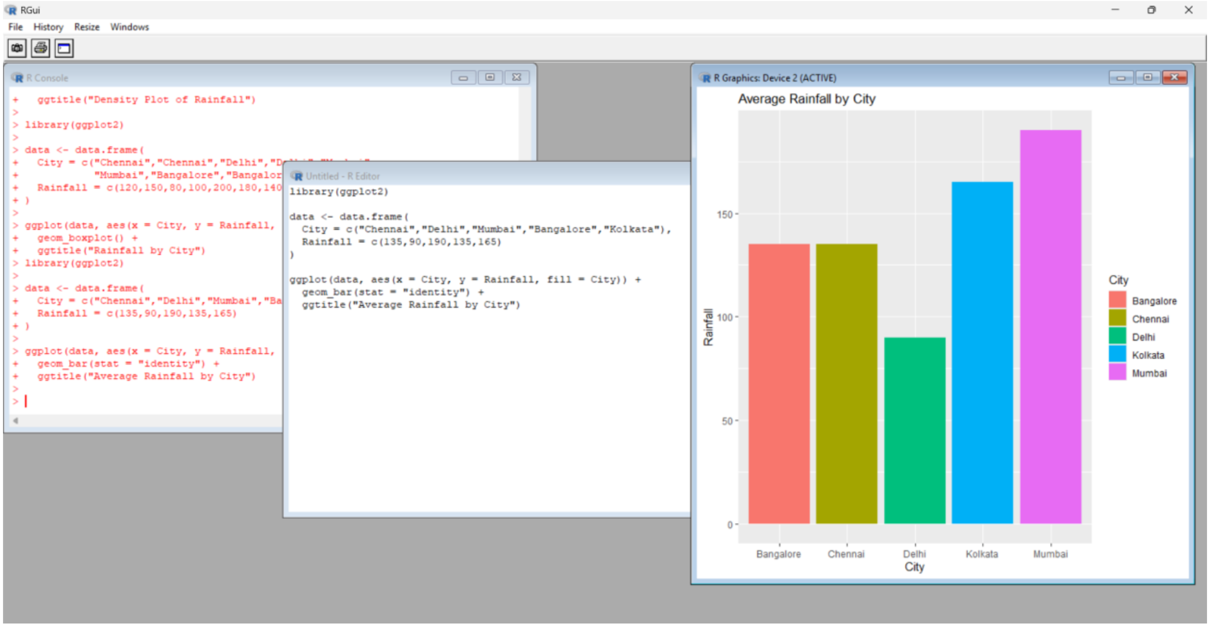


4.3. BOX PLOT

```
LIBRARY(GGPLOT2)

DATA <- DATA.FRAME(
  CITY = C("CHENNAI","CHENNAI","DELHI","DELHI","MUMBAI",
           "MUMBAI","BANGALORE","BANGALORE","KOLKATA","KOLKATA"),
  RAINFALL = C(120,150,80,100,200,180,140,130,170,160)
)

GGPLOT(DATA, AES(X = CITY, Y = RAINFALL, FILL = CITY)) +
  GEOM_BOXPLOT() +
  GGTITLE("RAINFALL BY CITY")
```

4.4. BAR CHART

```
LIBRARY(GGPLOT2)

DATA <- DATA.FRAME(
  CITY = C("CHENNAI","DELHI","MUMBAI","BANGALORE","KOLKATA"),
  RAINFALL = C(135,90,190,135,165)
)

GGPLOT(DATA, AES(X = CITY, Y = RAINFALL, FILL = CITY)) +
  GEOM_BAR(STAT = "IDENTITY") +
  GGTITLE("AVERAGE RAINFALL BY CITY")
```


RGui
File History Resize Windows
R Console
+ ggtitle("Density Plot of Rainfall")
> library(ggplot2)
> data <- data.frame(
> ggplot(data, aes(x = City, y = Rainfall,
+ geom_boxplot() +
+ ggtitle("Rainfall by City")
> library(ggplot2)
> data <- data.frame(
+ Rainfall = c(135,90,190,135,165)
> ggplot(data, aes(x = City, y = Rainfall,
+ geom_bar(stat = "identity") +
+ ggtitle("Average Rainfall by City")
Untitled - R Editor
library(ggplot2)
data <- data.frame(
City = c("Chennai","Delhi","Mumbai","Bangalore","Kolkata"),
Rainfall = c(135,90,190,135,165)
)
ggplot(data, aes(x = City, y = Rainfall, fill = City)) +
geom_bar(stat = "identity") +
ggtitle("Average Rainfall by City")
R Graphics: Device 2 (ACTIVE)
Average Rainfall by City
Rainfall
150
100
50
0
Bangalore
Chennai
Delhi
Kolkata
Mumbai
City
City
Bangalore
Chennai
Delhi
Kolkata
Mumbai

5. BANK LOAN RISK ASSESSMENT

A FINANCIAL INSTITUTION HAS CUSTOMER LOAN AMOUNTS, REPAYMENT HISTORY, INCOME LEVELS, AND CREDIT SCORES. THE ANALYSIS NEEDS TO SHOW LOAN DISTRIBUTIONS AND CUSTOMER SEGMENTS WHILE HIGHLIGHTING RISK PATTERNS, SKEWNESS, AND OUTLIERS FOR LOAN APPROVAL DECISIONS.

EVALUATION OF VISUALIZATIONS

VISUALIZATION	EVALUATION AND EVIDENCE
HISTOGRAM	USEFUL FOR IDENTIFYING SKEWNESS IN LOAN AMOUNTS AND INCOME LEVELS AND UNDERSTANDING THE FREQUENCY OF DIFFERENT FINANCIAL RANGES.
BOX PLOT	HIGHLIGHTS OUTLIERS IN LOAN AMOUNTS, INCOME, AND CREDIT SCORES AND PROVIDES A CLEAR COMPARISON OF RISK-RELATED VARIABLES ACROSS CUSTOMER GROUPS.
SCATTER PLOT	SHOWS RELATIONSHIPS SUCH AS INCOME VERSUS LOAN AMOUNT OR CREDIT SCORE VERSUS REPAYMENT PERFORMANCE. IT CAN REVEAL CLUSTERS AND POTENTIAL RISK PATTERNS.
BAR CHART	USEFUL FOR COMPARING LOAN APPROVAL OR DEFAULT RATES ACROSS CUSTOMER SEGMENTS, CREDIT-SCORE CATEGORIES, OR REPAYMENT GROUPS.
HEATMAP	ALTHOUGH NOT LISTED IN THE SCENARIO, IT CAN BE VALUABLE FOR SHOWING CORRELATIONS AMONG INCOME, LOAN AMOUNT, CREDIT SCORE, AND REPAYMENT VARIABLES.

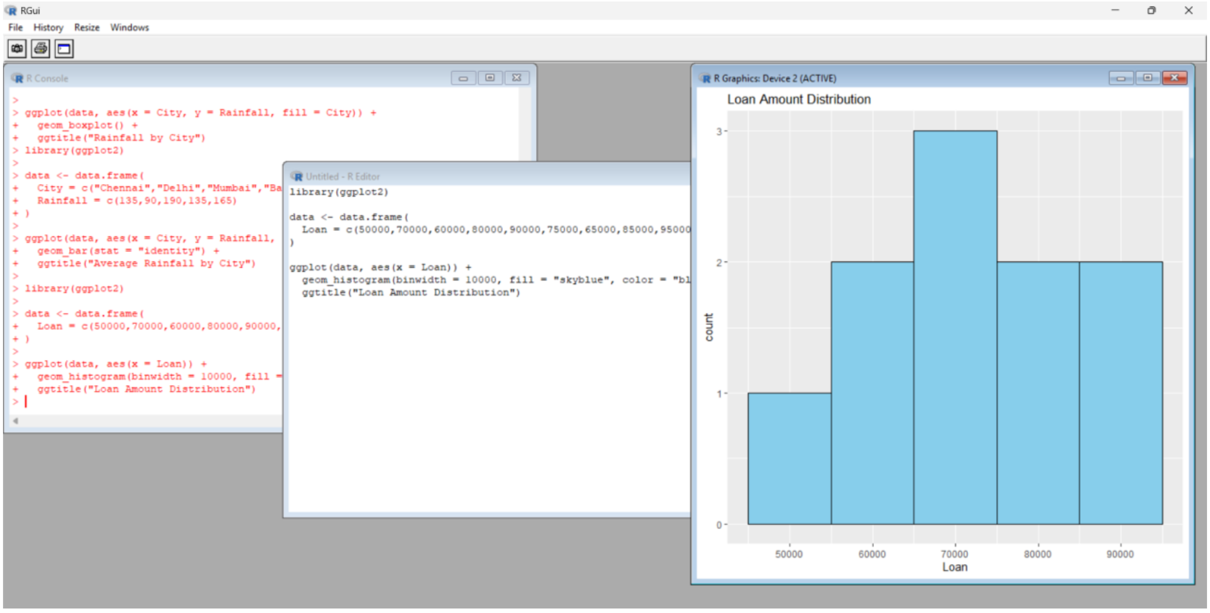
RECOMMENDATION: A **BOX PLOT** SHOULD BE INCLUDED IN THE MANAGEMENT DASHBOARD TO HIGHLIGHT SKEWNESS, VARIABILITY, AND OUTLIERS IN FINANCIAL VARIABLES. IT SHOULD BE SUPPORTED BY A **SCATTER PLOT** FOR RISK RELATIONSHIPS AND A **BAR CHART** FOR APPROVAL/DEFAULT RATES ACROSS CUSTOMER SEGMENTS. THIS COMBINATION PROVIDES BOTH STATISTICAL AND DECISION-ORIENTED INSIGHTS.

SUGGESTED LIBRARIES: PYTHON: SEABORN, PLOTLY, PANDAS **|** R: GGPLOT2

5.1. HISTOGRAM

LIBRARY(GGPLOT2)

```
DATA <- DATA.FRAME(
  LOAN = C(50000,70000,60000,80000,90000,75000,65000,85000,95000,70000)
)

GGPLOT(DATA, AES(X = LOAN)) +
  GEOM_HISTOGRAM(BINWIDTH = 10000, FILL = "SKYBLUE", COLOR = "BLACK") +
  GGTITLE("LOAN AMOUNT DISTRIBUTION")
```

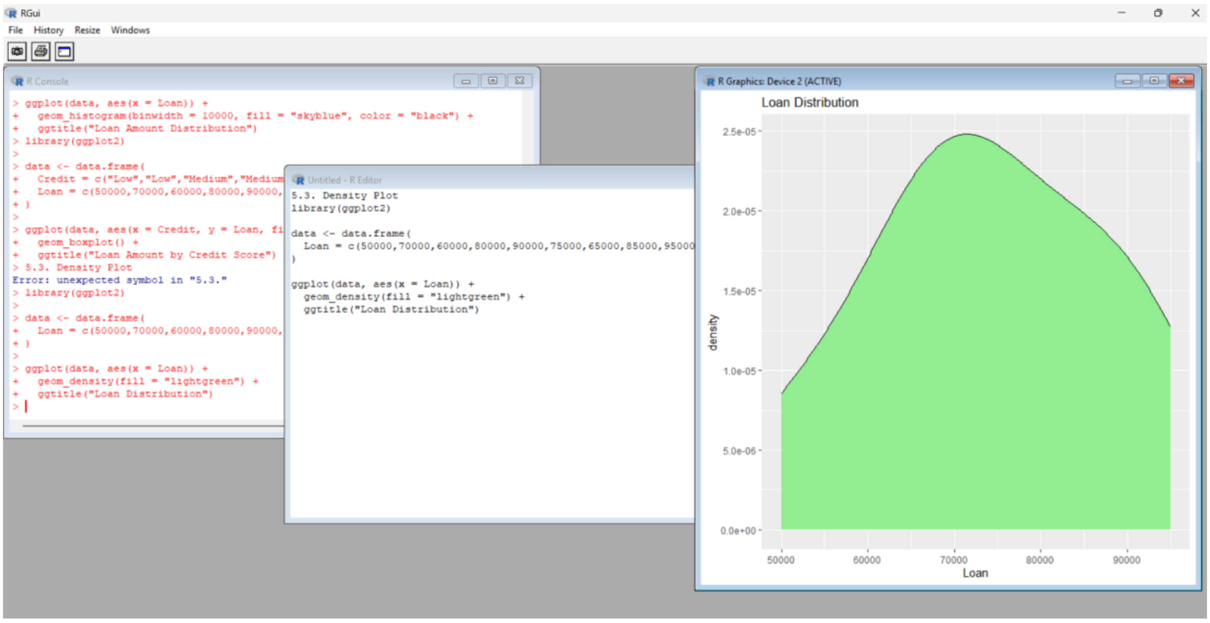


5.2. BOX PLOT

```
LIBRARY(GGPLOT2)

DATA <- DATA.FRAME(
  CREDIT =
C("LOW","LOW","MEDIUM","MEDIUM","HIGH","HIGH","LOW","MEDIUM","HIGH","HIG
H"),
  LOAN = C(50000,70000,60000,80000,90000,75000,65000,85000,95000,70000)
)

GGPLOT(DATA, AES(X = CREDIT, Y = LOAN, FILL = CREDIT)) +
  GEOM_BOXPLOT() +
  GGTITLE("LOAN AMOUNT BY CREDIT SCORE")
```

5.3. DENSITY PLOT

```
LIBRARY(GGPLOT2)

DATA <- DATA.FRAME(
  LOAN = C(50000,70000,60000,80000,90000,75000,65000,85000,95000,70000)
)

GGPLOT(DATA, AES(X = LOAN)) +
  GEOM_DENSITY(FILL = "LIGHTGREEN") +
  GGTITLE("LOAN DISTRIBUTION")
```



5.4. BAR CHART

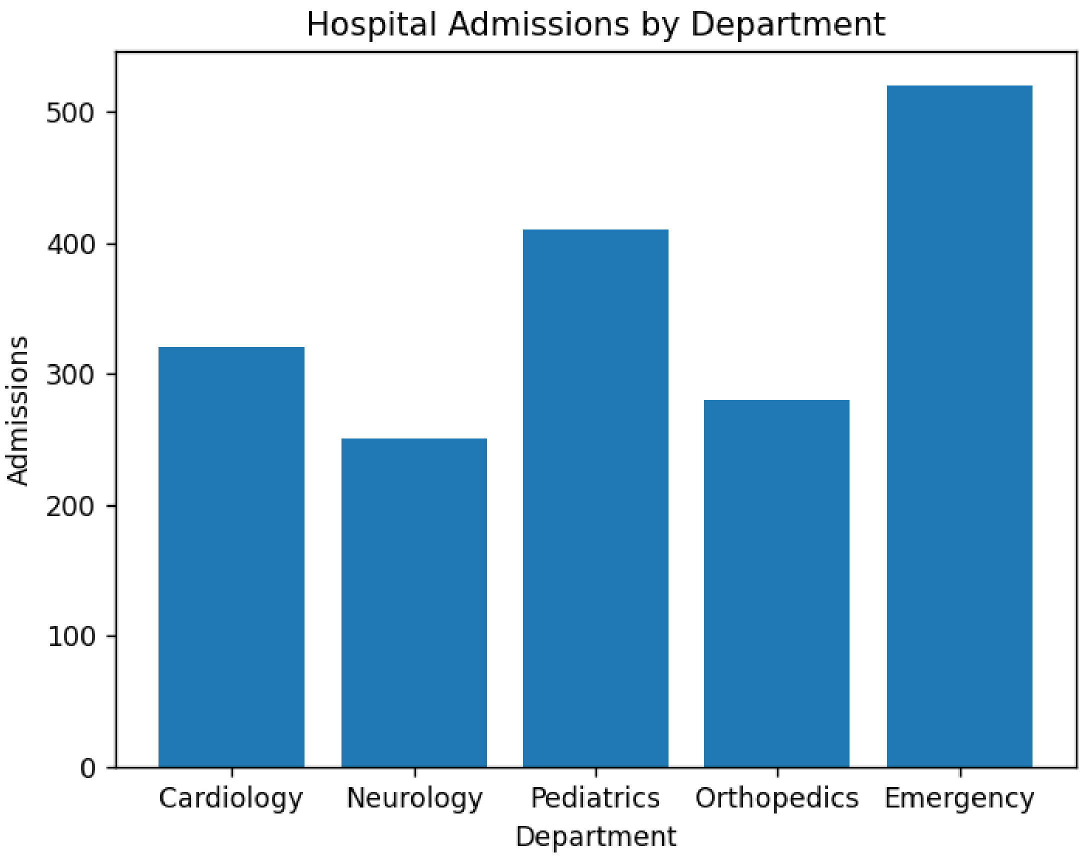
```
LIBRARY(GGPLOT2)

DATA <- DATA.FRAME(
  CREDIT = C("LOW","MEDIUM","HIGH"),
  CUSTOMERS = C(30,45,25)
)

GGPLOT(DATA, AES(X = CREDIT, Y = CUSTOMERS, FILL = CREDIT)) +
  GEOM_BAR(STAT = "IDENTITY") +
  GGTITLE("CUSTOMER SEGMENTS")
```

PYTHON CODES

1. HOSPITAL PATIENT ADMISSION ANALYSIS

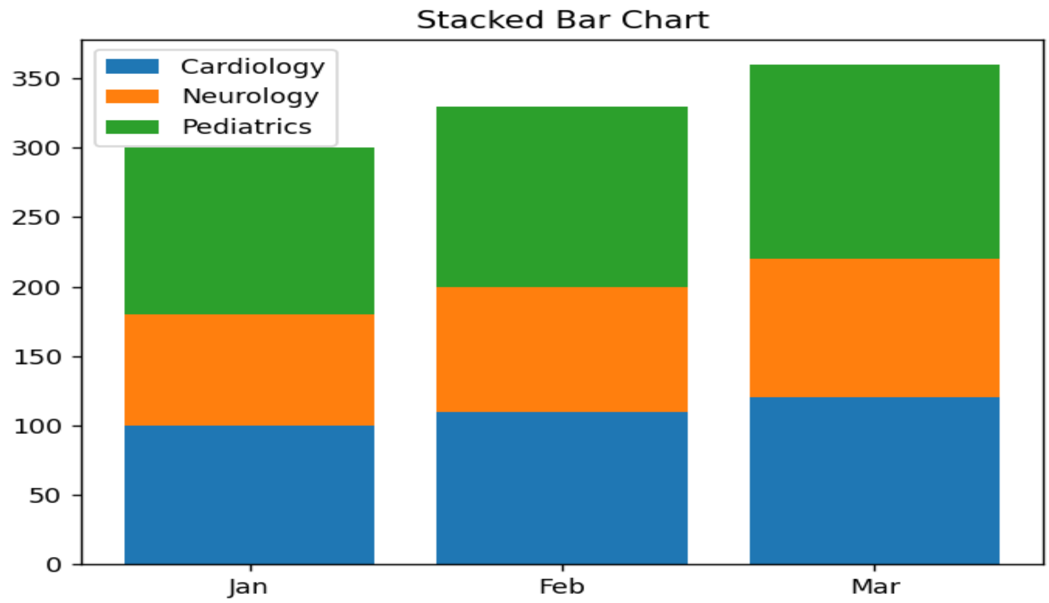
1.1. BAR CHART

```
IMPORT MATPLOTLIB.PYPLOT AS PLT

DEPARTMENTS = ["CARDIOLOGY", "NEUROLOGY", "PEDIATRICS", "ORTHOPEDICS",
"EMERGENCY"]
ADMISSIONS = [320, 250, 410, 280, 520]

PLT.BAR(DEPARTMENTS, ADMISSIONS)
PLT.TITLE("HOSPITAL ADMISSIONS BY DEPARTMENT")
PLT.XLABEL("DEPARTMENT")
PLT.YLABEL("ADMISSIONS")
PLT.SHOW()
```



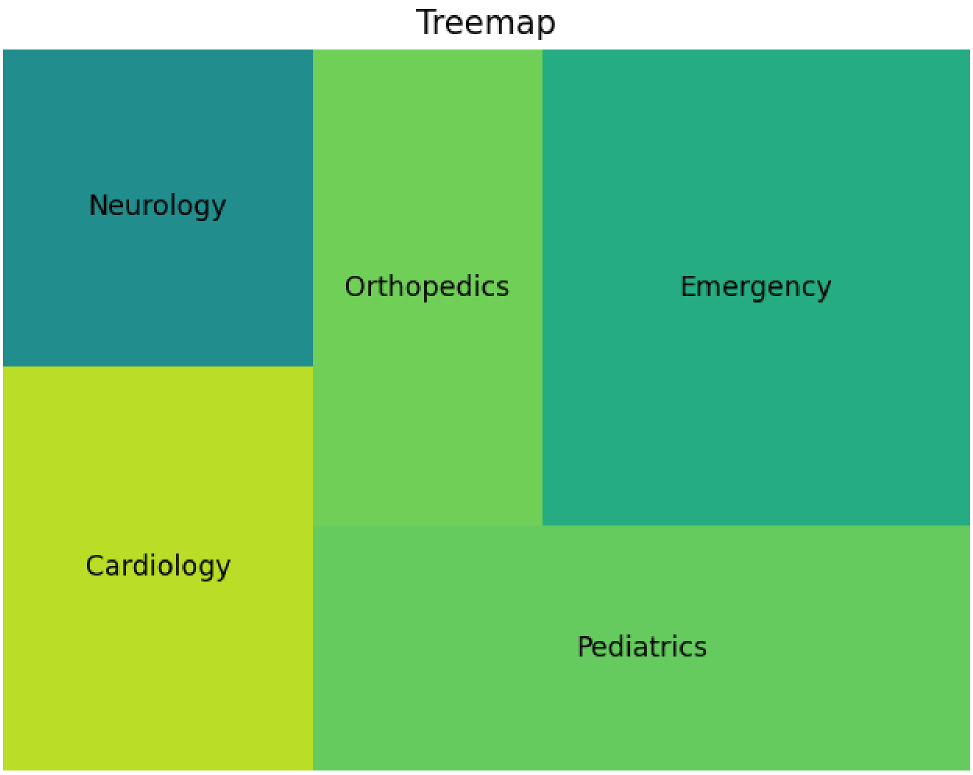
1.2. STACKED BAR CHART

```
IMPORT MATPLOTLIB.PYPLOT AS PLT

MONTHS = ["JAN", "FEB", "MAR"]

CARDIOLOGY = [100, 110, 120]
NEUROLOGY = [80, 90, 100]
PEDIATRICS = [120, 130, 140]

PLT.BAR(MONTHS, CARDIOLOGY, LABEL="CARDIOLOGY")
PLT.BAR(MONTHS, NEUROLOGY, BOTTOM=CARDIOLOGY, LABEL="NEUROLOGY")

BOTTOM = [CARDIOLOGY[I] + NEUROLOGY[I] FOR I IN RANGE(3)]
PLT.BAR(MONTHS, PEDIATRICS, BOTTOM=BOTTOM, LABEL="PEDIATRICS")

PLT.TITLE("STACKED BAR CHART")
PLT.LEGEND()
PLT.SHOW()
```



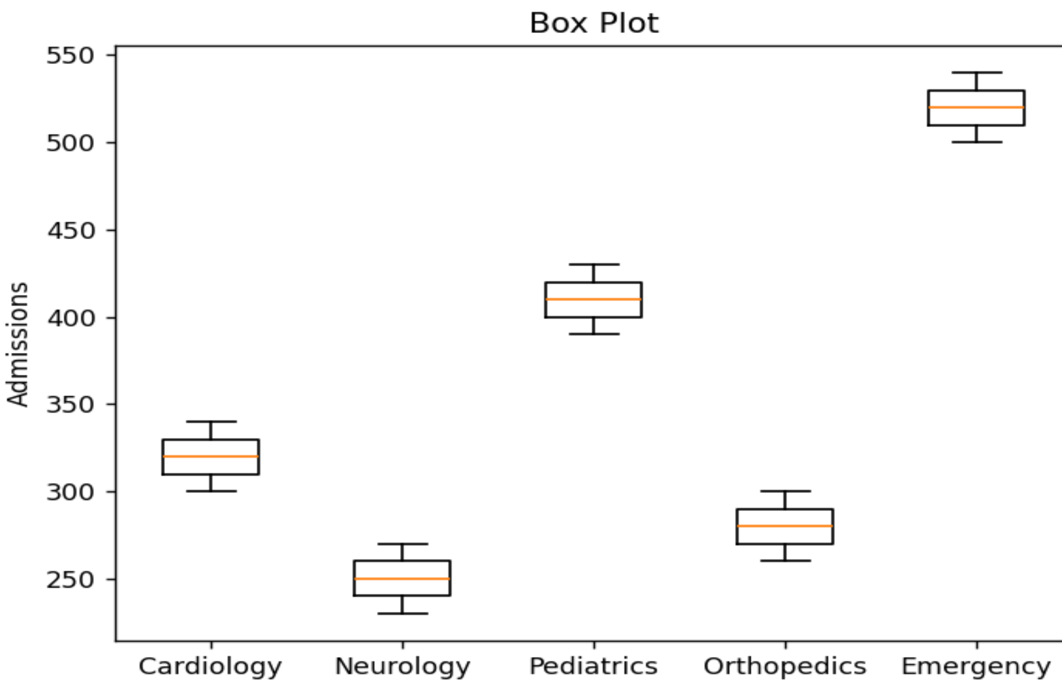
1.3. TREEMAP

```
IMPORT MATPLOTLIB.PYPLOT AS PLT
IMPORT SQUARIFY

LABELS = ["CARDIOLOGY", "NEUROLOGY", "PEDIATRICS", "ORTHOPEDICS",
"EMERGENCY"]
SIZES = [320, 250, 410, 280, 520]

SQUARIFY.PLOT(SIZES=SIZES, LABEL=LABELS)
PLT.TITLE("TREEMAP")
PLT.AXIS("OFF")
PLT.SHOW()
```



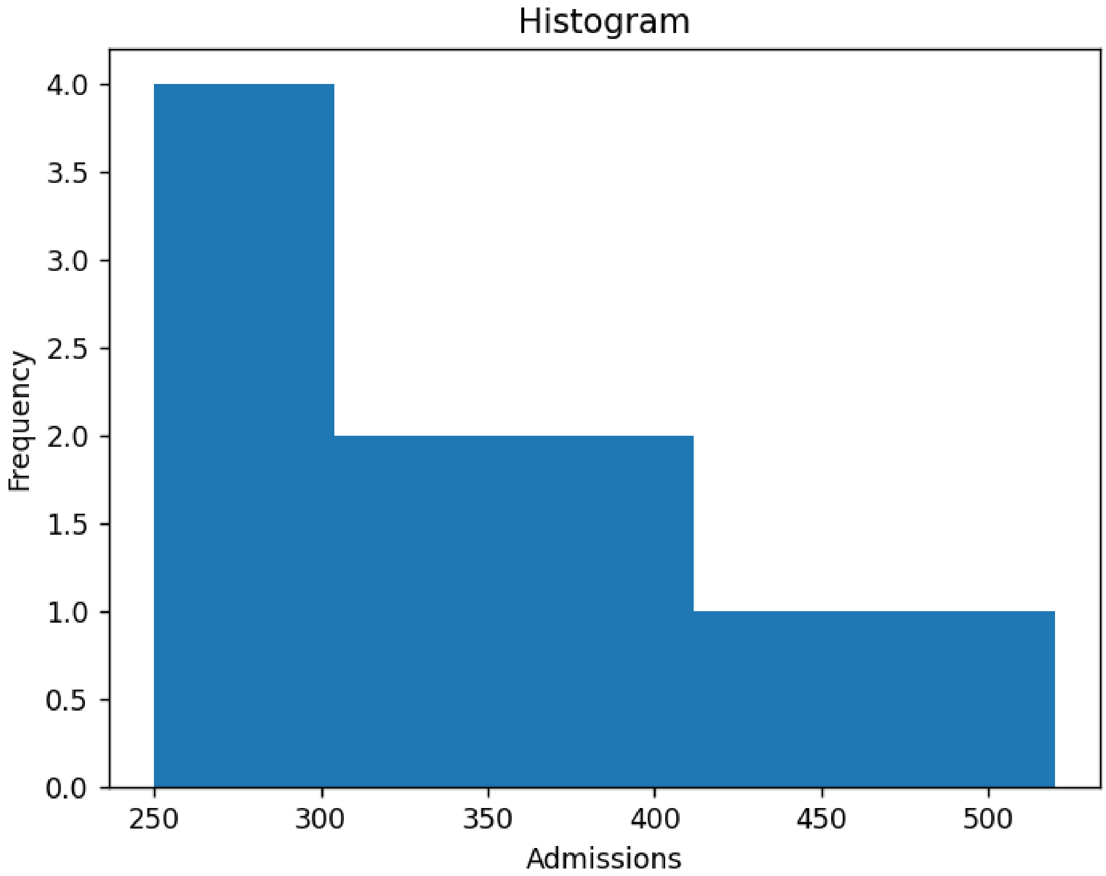
1.4. BOX PLOT

```
IMPORT MATPLOTLIB.PYPLOT AS PLT

DATA = [
    [300, 320, 340],
    [230, 250, 270],
    [390, 410, 430],
    [260, 280, 300],
    [500, 520, 540]
]

LABELS = ["CARDIOLOGY", "NEUROLOGY", "PEDIATRICS", "ORTHOPEDICS",
"EMERGENCY"]

PLT.BOXPLOT(DATA, LABELS=LABELS)
PLT.TITLE("BOX PLOT")
PLT.YLABEL("ADMISSIONS")
PLT.SHOW()
```



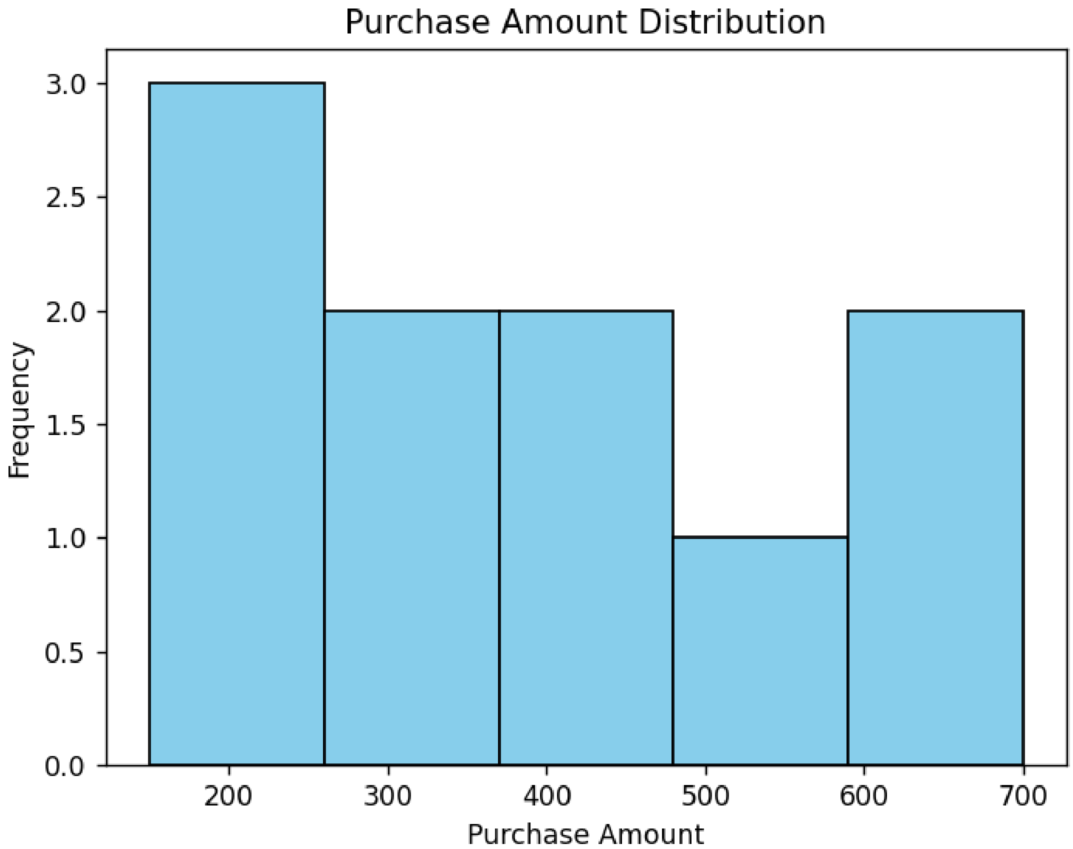
1.5. HISTOGRAM

```
IMPORT MATPLOTLIB.PYPLOT AS PLT

ADMISSIONS = [320, 250, 410, 280, 520, 300, 340, 270, 390, 430]

PLT.HIST(ADMISSIONS, BINS=5)
PLT.TITLE("HISTOGRAM")
PLT.XLABEL("ADMISSIONS")
PLT.YLABEL("FREQUENCY")
PLT.SHOW()
```



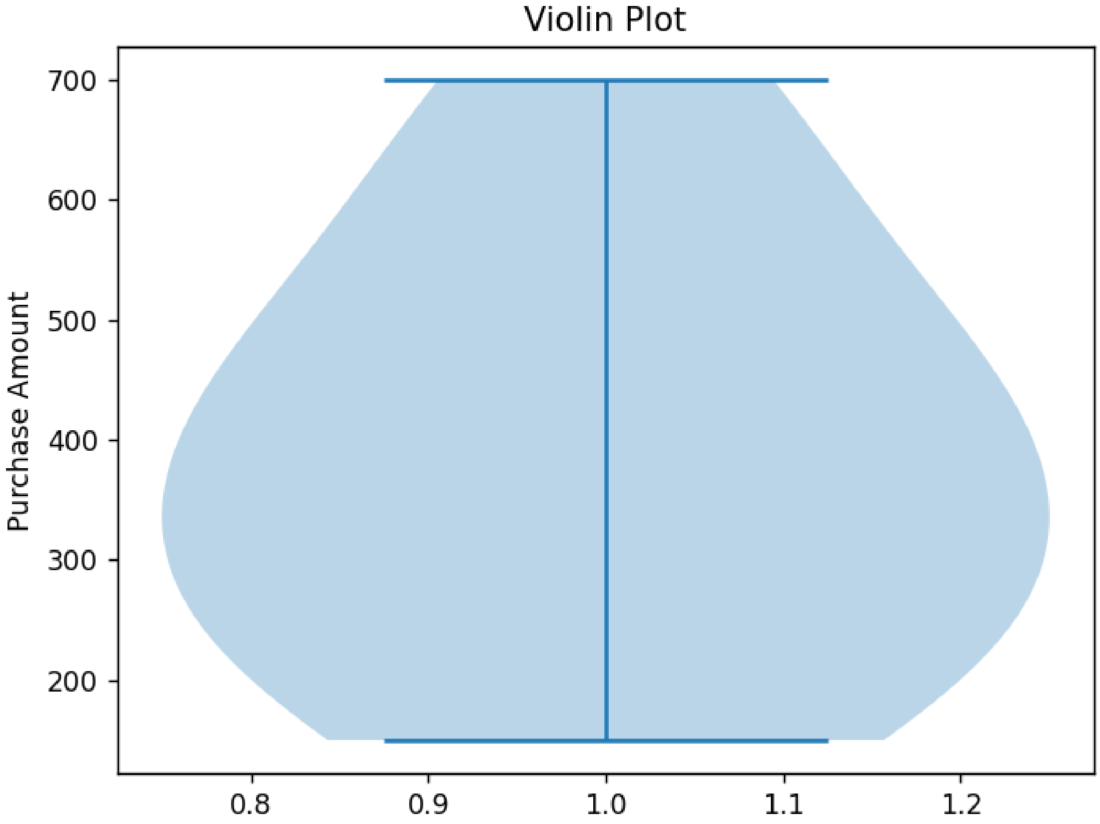
2. E-COMMERCE CUSTOMER SPENDING BEHAVIOUR
2.1. HISTOGRAM

```
IMPORT MATPLOTLIB.PYPLOT AS PLT

PURCHASE = [200,350,150,500,450,300,250,600,700,400]

PLT.HIST(PURCHASE, BINS=5, COLOR="SKYBLUE", EDGECOLOR="BLACK")
PLT.TITLE("PURCHASE AMOUNT DISTRIBUTION")
PLT.XLABEL("PURCHASE AMOUNT")
PLT.YLABEL("FREQUENCY")
PLT.SHOW()
```



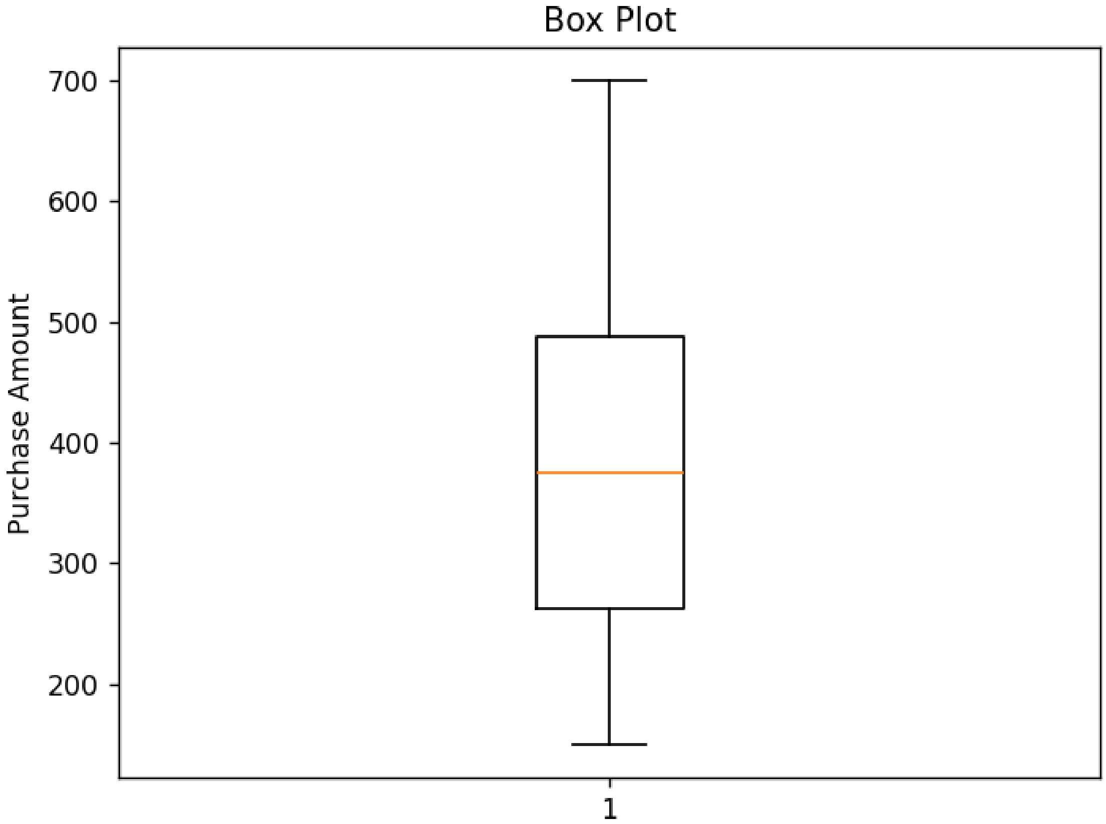
2.2. VIOLIN PLOT

```
IMPORT MATPLOTLIB.PYPLOT AS PLT

PURCHASE = [200,350,150,500,450,300,250,600,700,400]

PLT.VIOLINPLOT(PURCHASE)
PLT.TITLE("VIOLIN PLOT")
PLT.YLABEL("PURCHASE AMOUNT")
PLT.SHOW()
```



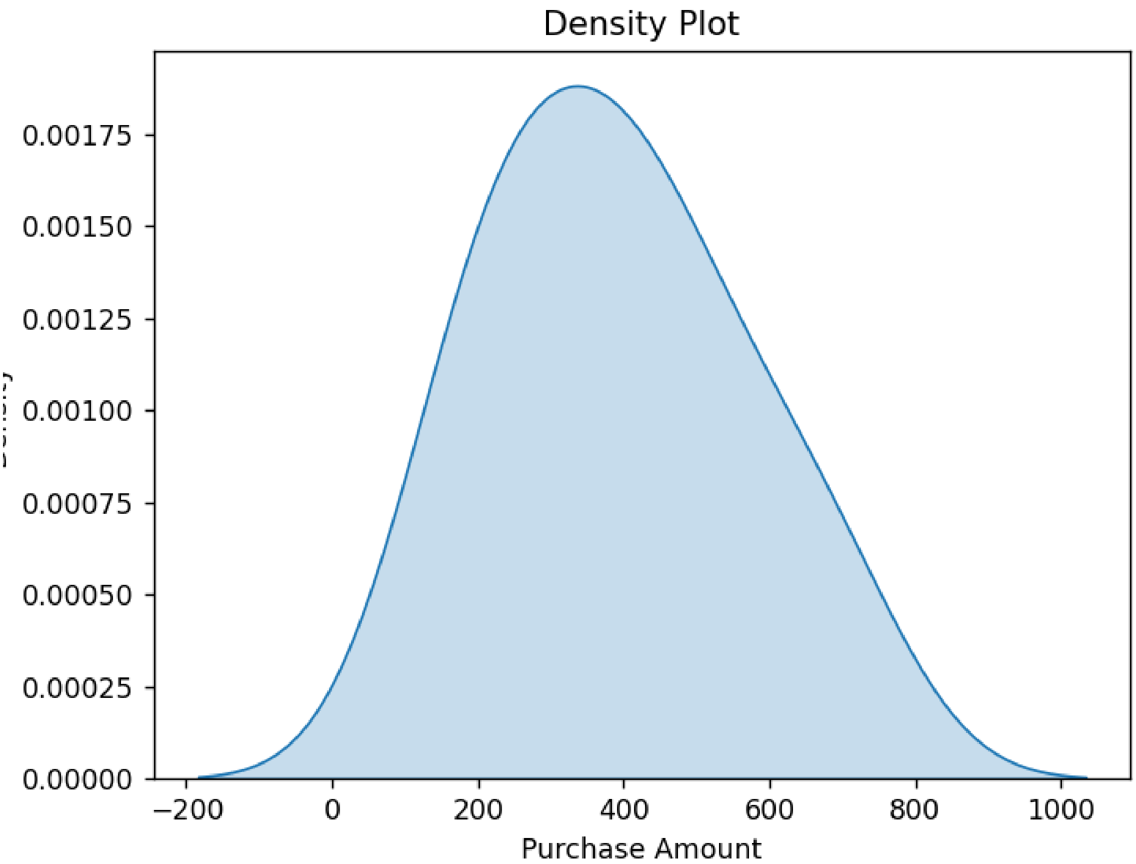
2.3. BOX PLOT

```
IMPORT MATPLOTLIB.PYPLOT AS PLT

PURCHASE = [200,350,150,500,450,300,250,600,700,400]

PLT.BOXPLOT(PURCHASE)
PLT.TITLE("BOX PLOT")
PLT.YLABEL("PURCHASE AMOUNT")
PLT.SHOW()
```



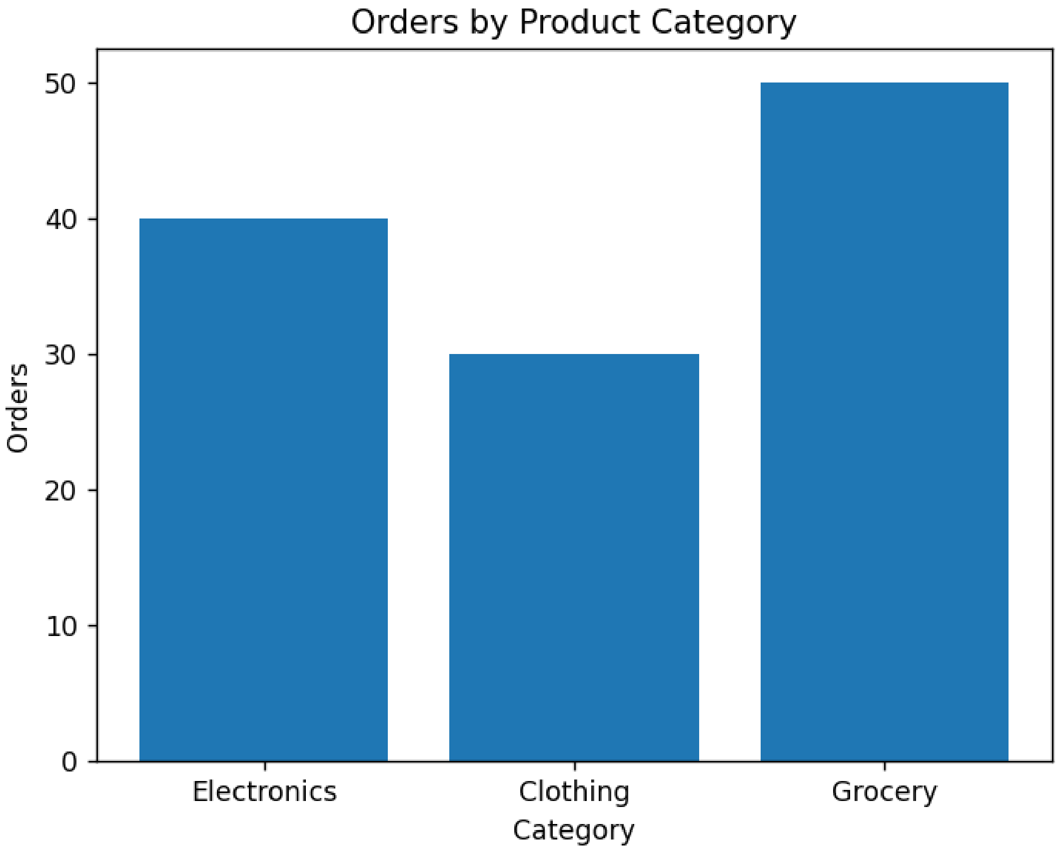
2.4. DENSITY PLOT

```
IMPORT SEABORN AS SNS
IMPORT MATPLOTLIB.PYPLOT AS PLT

PURCHASE = [200,350,150,500,450,300,250,600,700,400]

SNS.KDEPLOT(PURCHASE, FILL=TRUE)
PLT.TITLE("DENSITY PLOT")
PLT.XLABEL("PURCHASE AMOUNT")
PLT.SHOW()
```



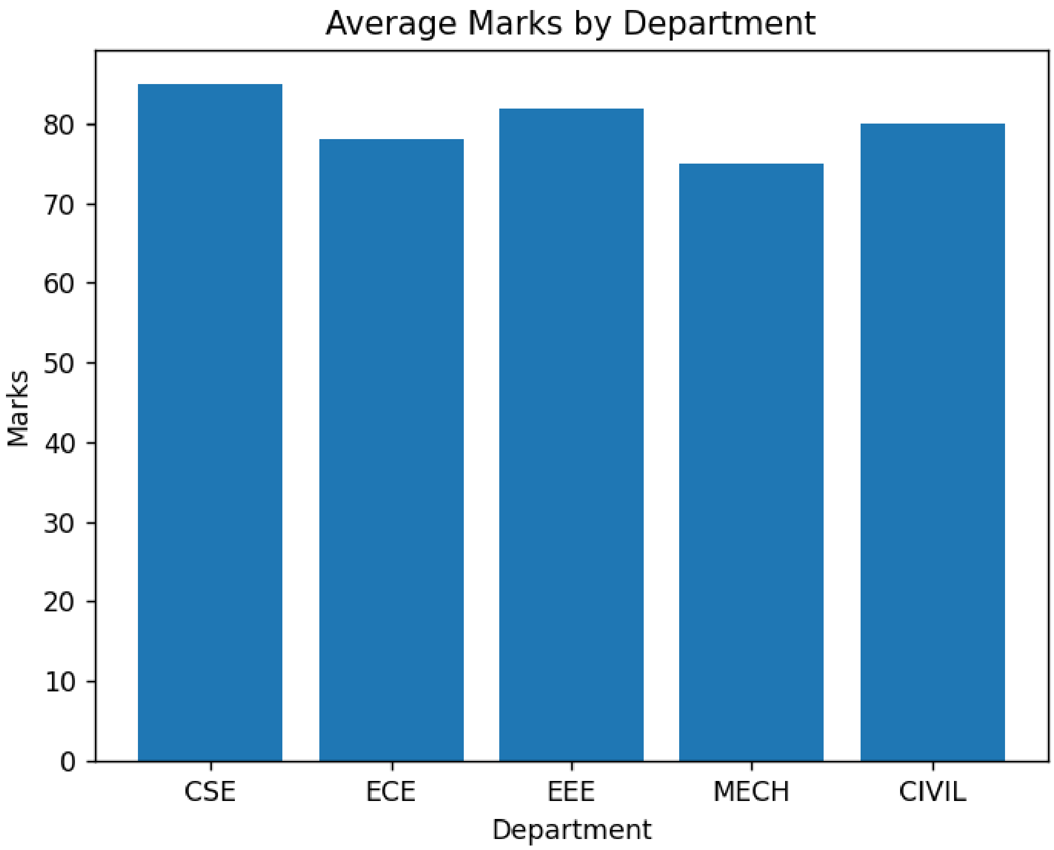
2.5. BAR CHART

```
IMPORT MATPLOTLIB.PYPLOT AS PLT

CATEGORY = ["ELECTRONICS","CLOTHING","GROCERY"]
ORDERS = [40,30,50]

PLT.BAR(CATEGORY, ORDERS)
PLT.TITLE("ORDERS BY PRODUCT CATEGORY")
PLT.XLABEL("CATEGORY")
PLT.YLABEL("ORDERS")
PLT.SHOW()
```



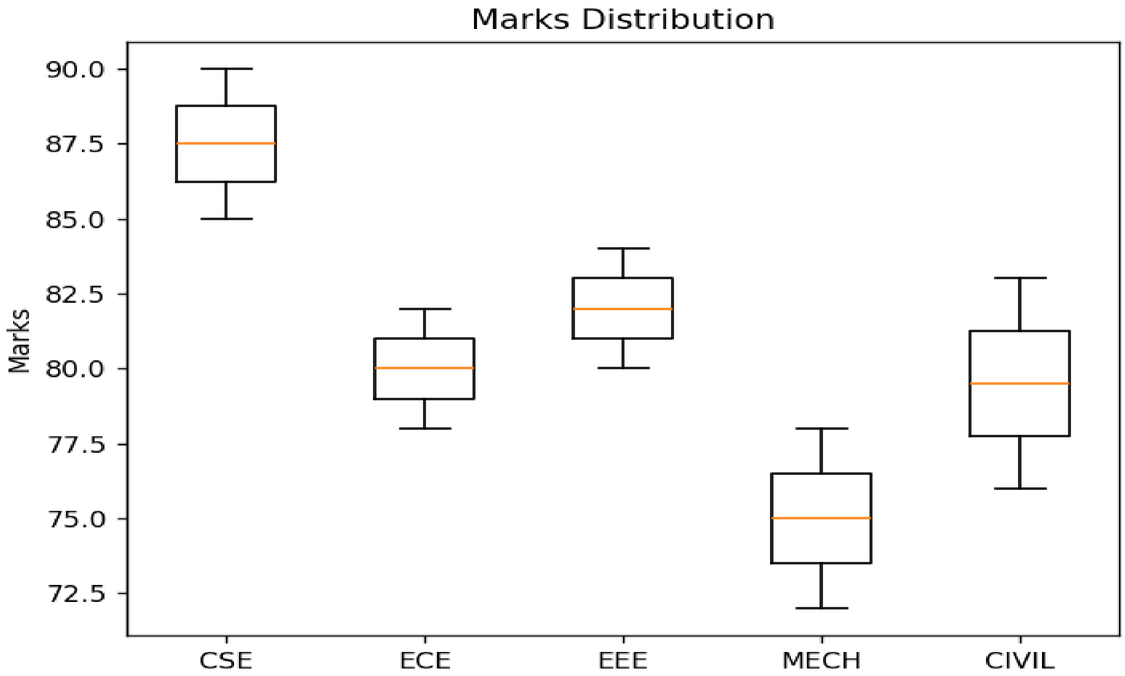
3. UNIVERSITY EXAMINATION PERFORMANCE DASHBOARD
3.1. BAR CHART

```
IMPORT MATPLOTLIB.PYPLOT AS PLT

DEPARTMENTS = ["CSE", "ECE", "EEE", "MECH", "CIVIL"]
MARKS = [85, 78, 82, 75, 80]

PLT.BAR(DEPARTMENTS, MARKS)
PLT.TITLE("AVERAGE MARKS BY DEPARTMENT")
PLT.XLABEL("DEPARTMENT")
PLT.YLABEL("MARKS")
PLT.SHOW()
```



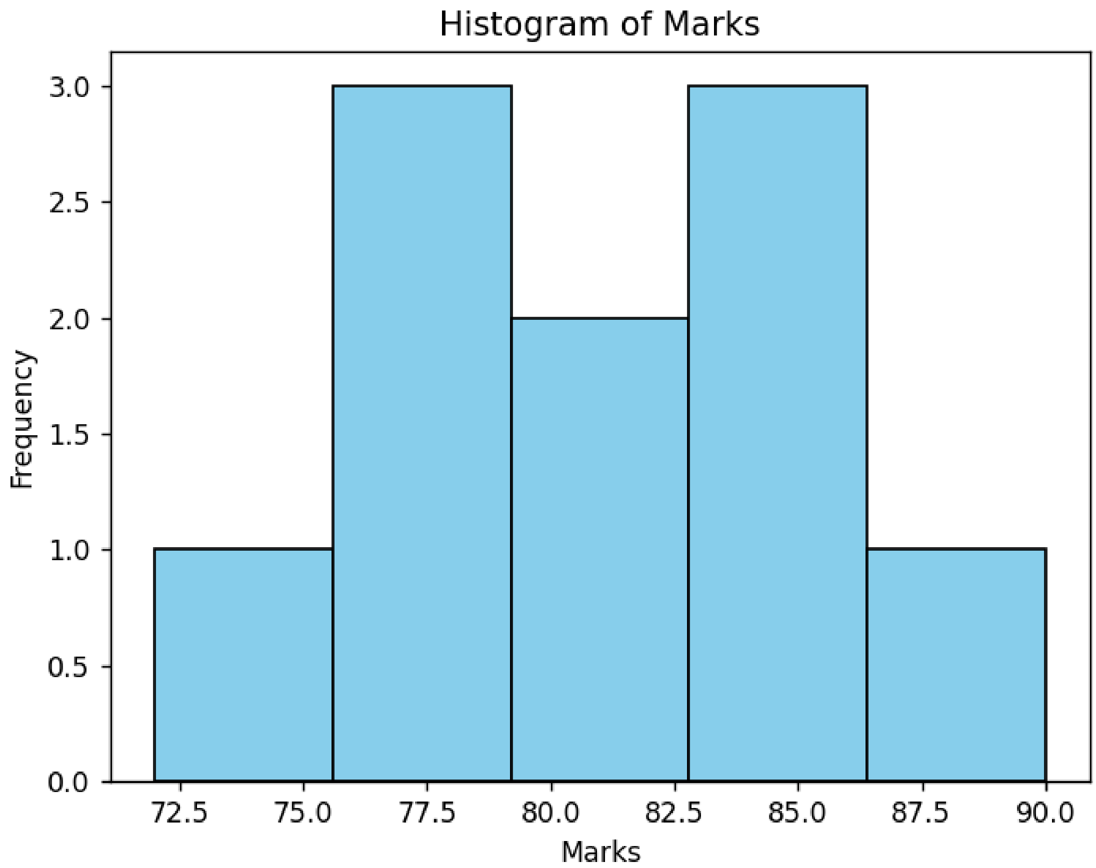
3.2. BOX PLOT

```
IMPORT MATPLOTLIB.PYPLOT AS PLT

DATA = [
    [85, 90],
    [78, 82],
    [80, 84],
    [72, 78],
    [76, 83]
]

LABELS = ["CSE", "ECE", "EEE", "MECH", "CIVIL"]

PLT.BOXPLOT(DATA, LABELS=LABELS)
PLT.TITLE("MARKS DISTRIBUTION")
PLT.YLABEL("MARKS")
PLT.SHOW()
```



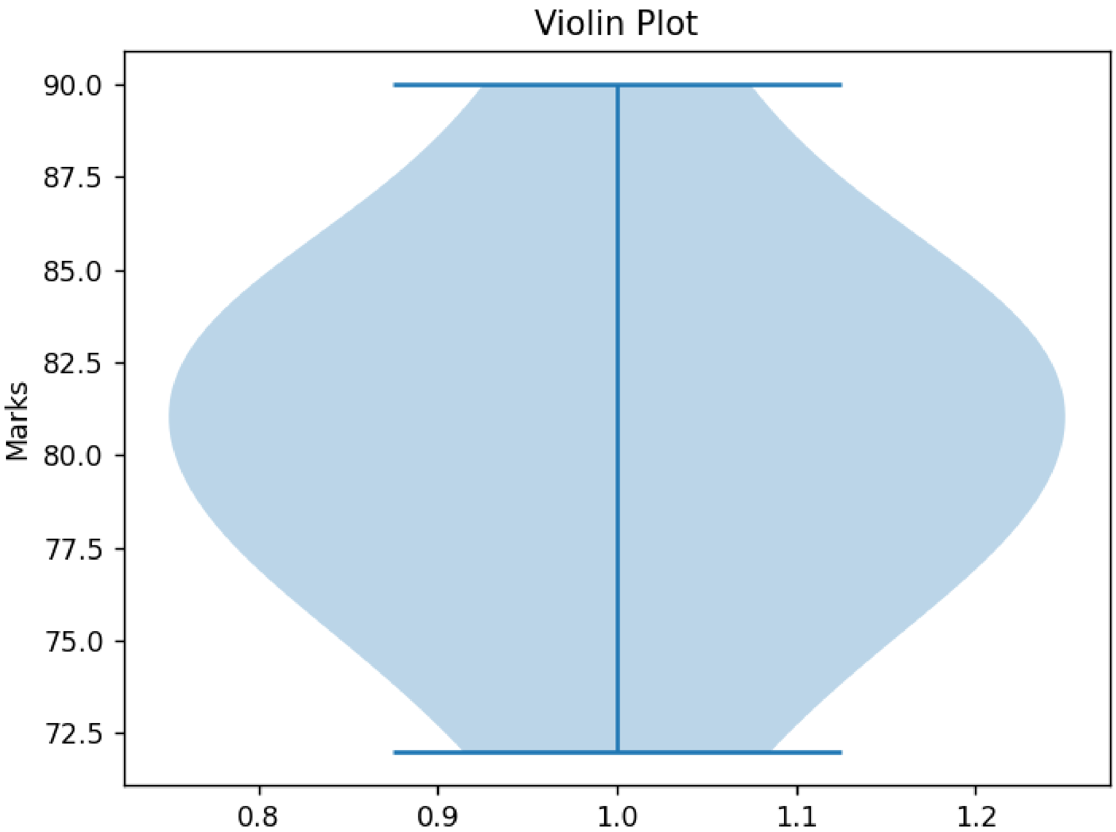
3.3. HISTOGRAM

```
IMPORT MATPLOTLIB.PYPLOT AS PLT

MARKS = [85, 90, 78, 82, 80, 84, 72, 78, 76, 83]

PLT.HIST(MARKS, BINS=5, COLOR="SKYBLUE", EDGECOLOR="BLACK")
PLT.TITLE("HISTOGRAM OF MARKS")
PLT.XLABEL("MARKS")
PLT.YLABEL("FREQUENCY")
PLT.SHOW()
```



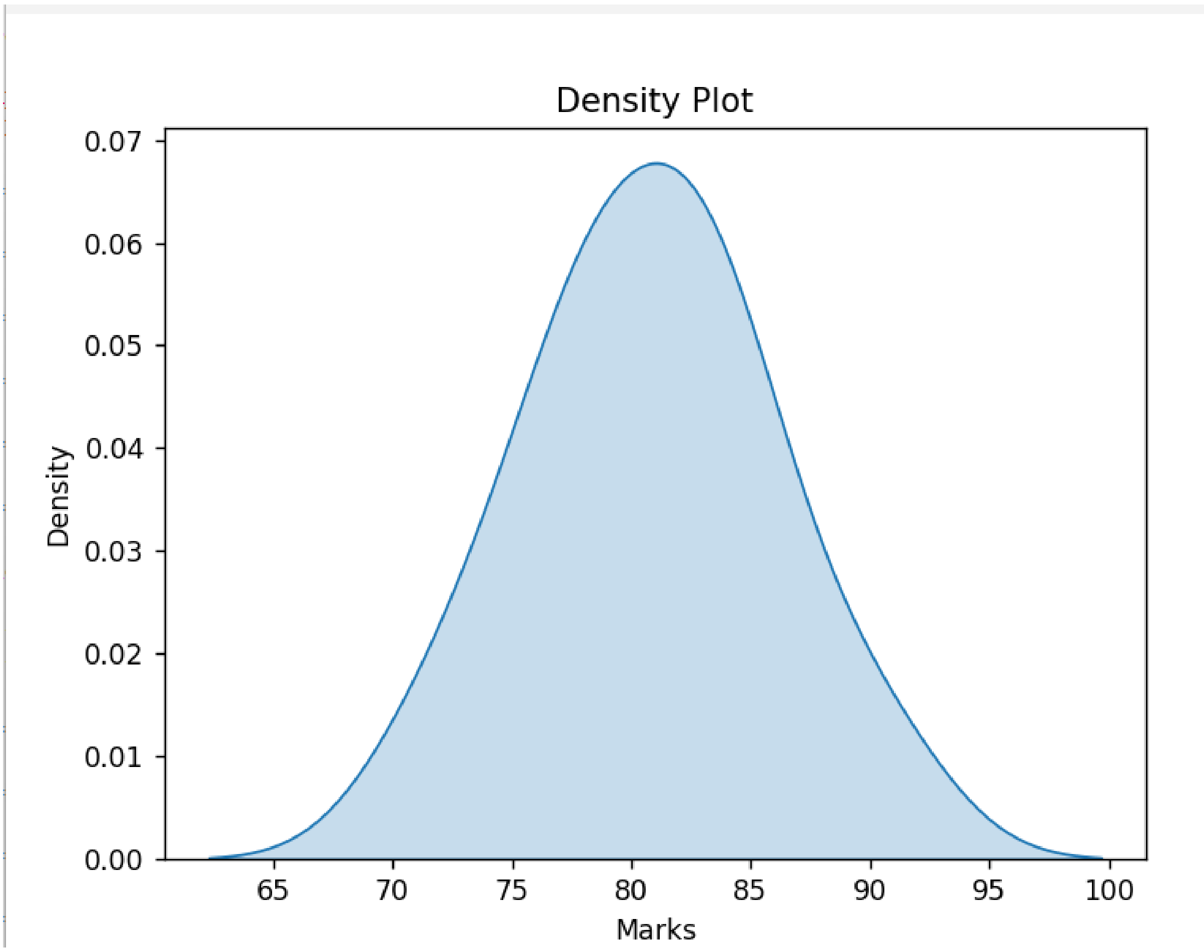
3.4. VIOLIN PLOT

```
IMPORT MATPLOTLIB.PYPLOT AS PLT

MARKS = [85, 90, 78, 82, 80, 84, 72, 78, 76, 83]

PLT.VIOLINPLOT(MARKS)
PLT.TITLE("VIOLIN PLOT")
PLT.YLABEL("MARKS")
PLT.SHOW()
```



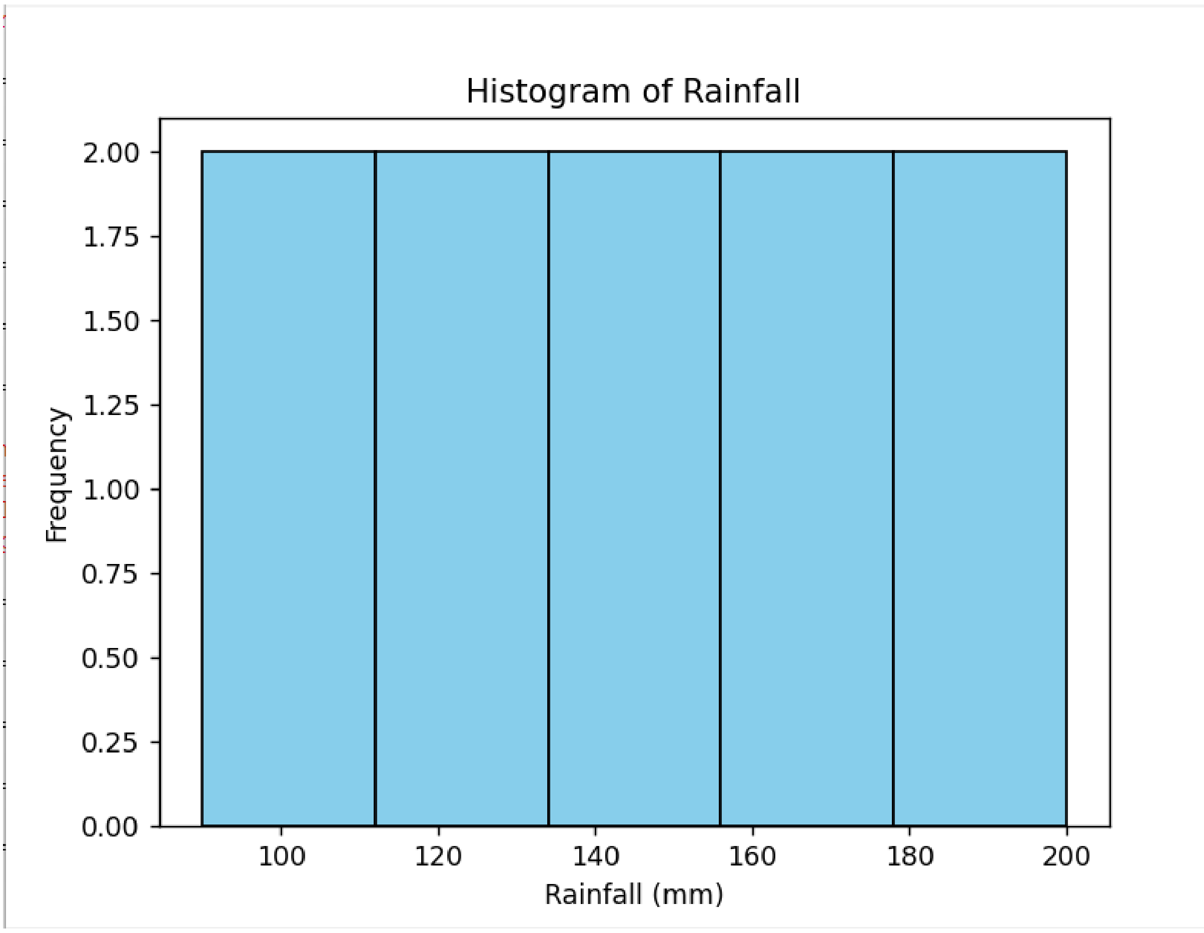
3.5. DENSITY PLOT

```
IMPORT SEABORN AS SNS
IMPORT MATPLOTLIB.PYPLOT AS PLT

MARKS = [85, 90, 78, 82, 80, 84, 72, 78, 76, 83]

SNS.KDEPLOT(MARKS, FILL=TRUE)
PLT.TITLE("DENSITY PLOT")
PLT.XLABEL("MARKS")
PLT.SHOW()
```



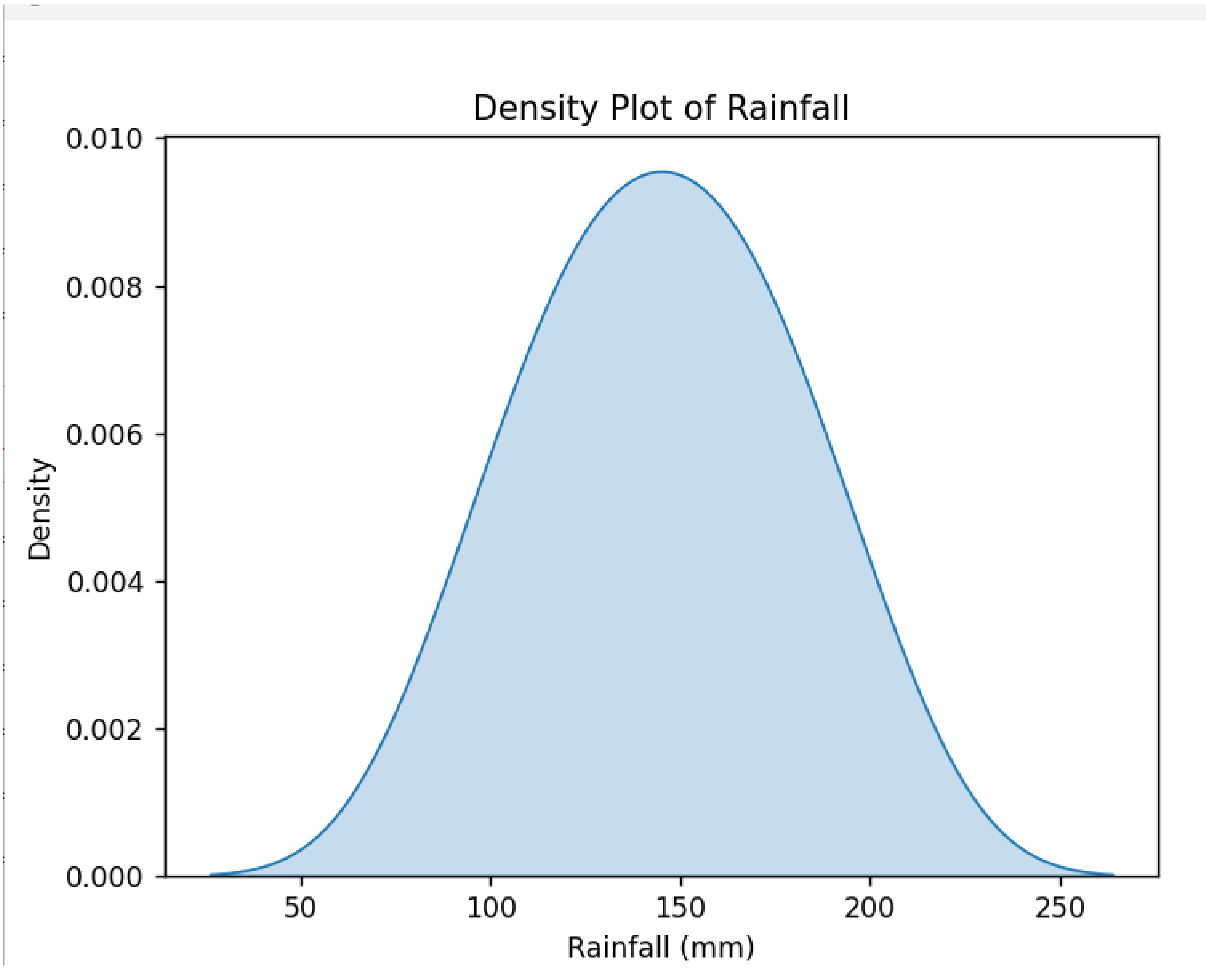
4.CLIMATE AND RAINFALL DISTRIBUTION STUDY
4.1. HISTOGRAM

```
IMPORT MATPLOTLIB.PYPLOT AS PLT

RAINFALL = [120,150,180,200,170,140,110,90,130,160]

PLT.HIST(RAINFALL, BINS=5, COLOR="SKYBLUE", EDGECOLOR="BLACK")
PLT.TITLE("HISTOGRAM OF RAINFALL")
PLT.XLABEL("RAINFALL (MM)")
PLT.YLABEL("FREQUENCY")
PLT.SHOW()
```



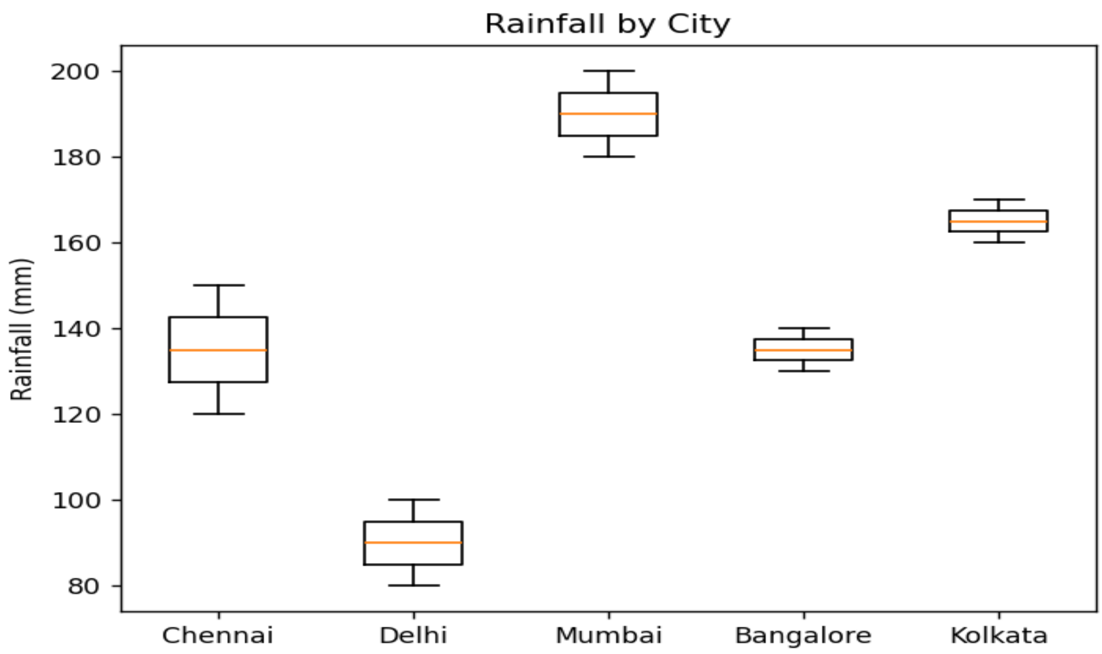
4.2. DENSITY PLOT

```
IMPORT SEABORN AS SNS
IMPORT MATPLOTLIB.PYPLOT AS PLT

RAINFALL = [120,150,180,200,170,140,110,90,130,160]

SNS.KDEPLOT(RAINFALL, FILL=TRUE)
PLT.TITLE("DENSITY PLOT OF RAINFALL")
PLT.XLABEL("RAINFALL (MM)")
PLT.SHOW()
```



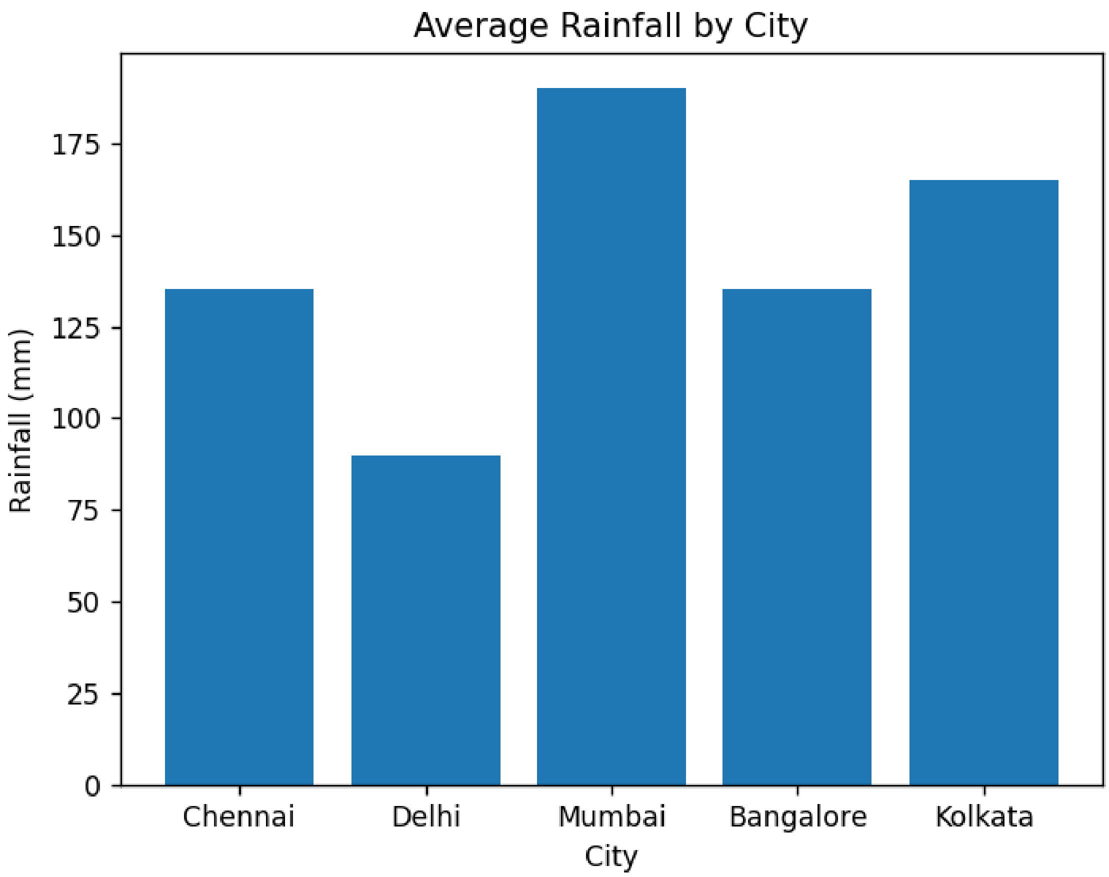
4.3. BOX PLOT

```
IMPORT MATPLOTLIB.PYPLOT AS PLT

DATA = [
    [120,150],
    [80,100],
    [200,180],
    [140,130],
    [170,160]
]

LABELS = ["CHENNAI","DELHI","MUMBAI","BANGALORE","KOLKATA"]

PLT.BOXPLOT(DATA, LABELS=LABELS)
PLT.TITLE("RAINFALL BY CITY")
PLT.YLABEL("RAINFALL (MM)")
PLT.SHOW()
```



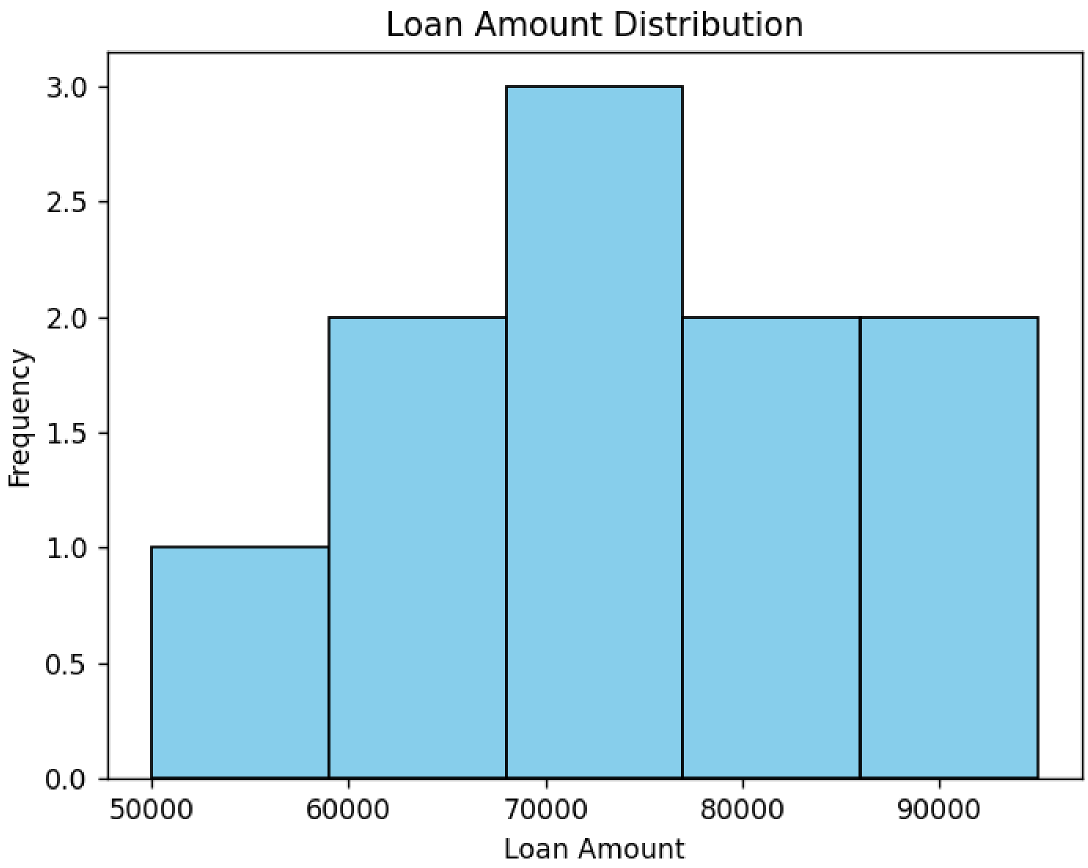
4.4. BAR CHART

```
IMPORT MATPLOTLIB.PYPLOT AS PLT

CITIES = ["CHENNAI","DELHI","MUMBAI","BANGALORE","KOLKATA"]
RAINFALL = [135,90,190,135,165]

PLT.BAR(CITIES, RAINFALL)
PLT.TITLE("AVERAGE RAINFALL BY CITY")
PLT.XLABEL("CITY")
PLT.YLABEL("RAINFALL (MM)")
PLT.SHOW()
```



5. BANK LOAN RISK ASSESSMENT
5.1. HISTOGRAM

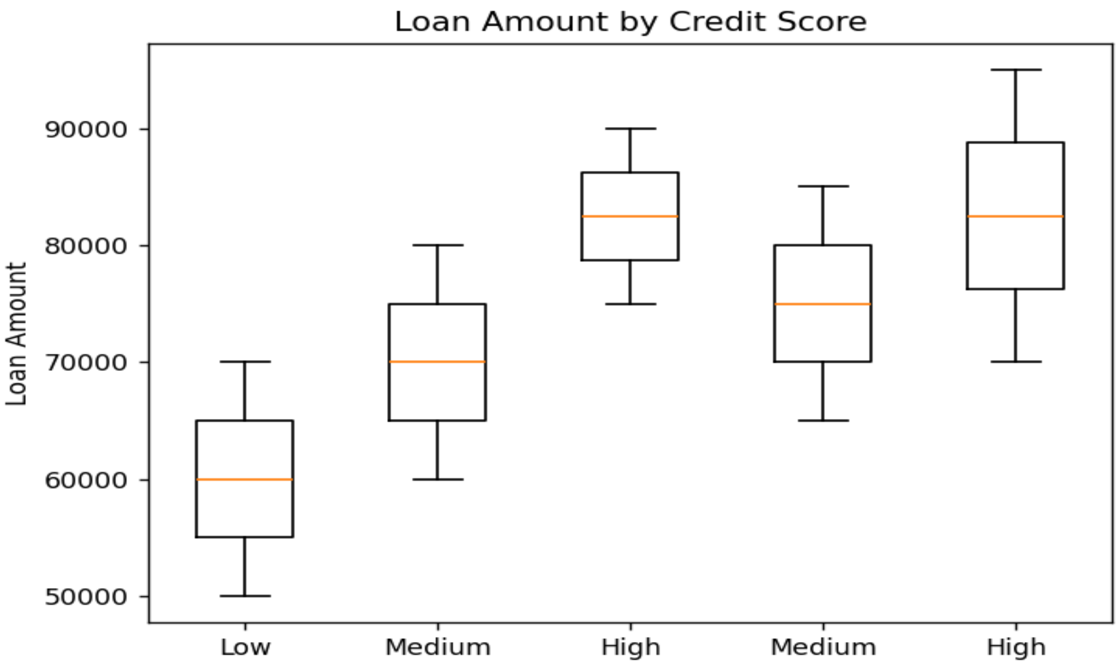
```
IMPORT MATPLOTLIB.PYPLOT AS PLT

LOAN = [50000,70000,60000,80000,90000,75000,65000,85000,95000,70000]

PLT.HIST(LOAN, BINS=5, COLOR="SKYBLUE", EDGECOLOR="BLACK")
PLT.TITLE("LOAN AMOUNT DISTRIBUTION")
PLT.XLABEL("LOAN AMOUNT")
PLT.YLABEL("FREQUENCY")
PLT.SHOW()
```



5.2. BOX PLOT

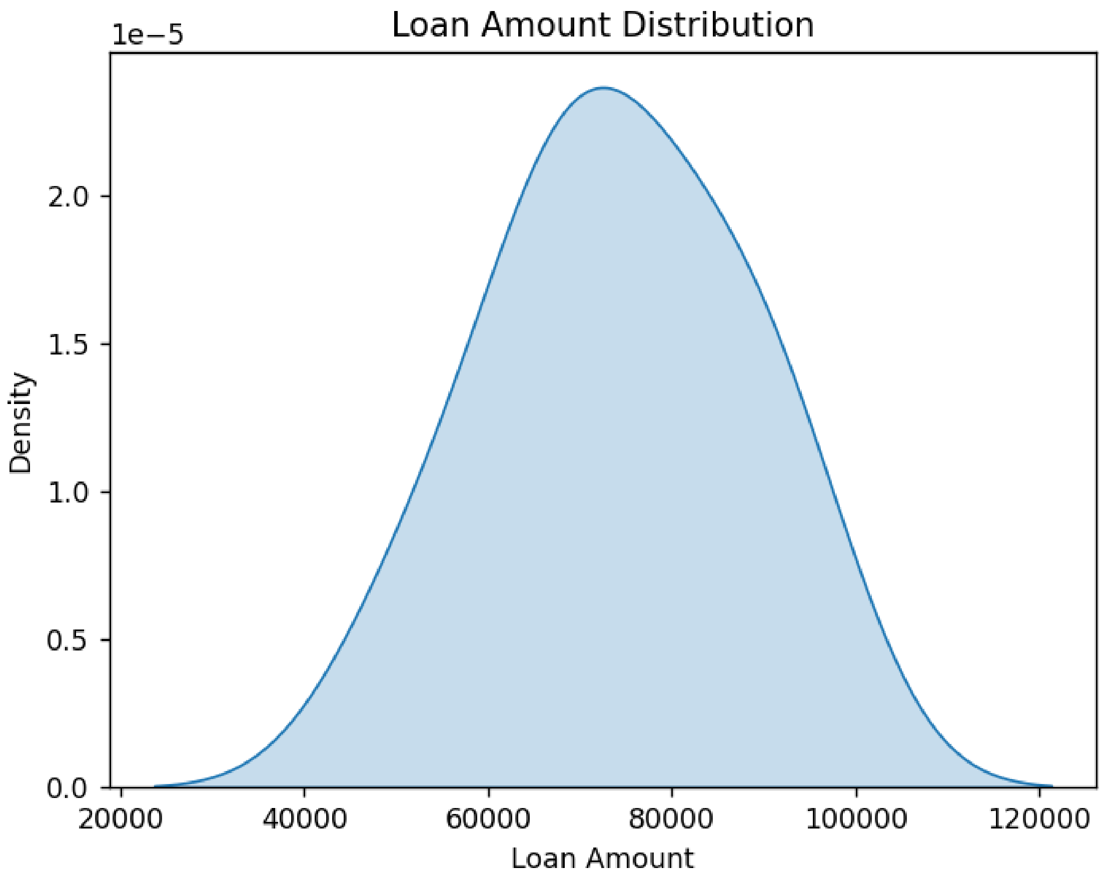
```
IMPORT MATPLOTLIB.PYPLOT AS PLT

DATA = [
    [50000,70000],
    [60000,80000],
    [90000,75000],
    [65000,85000],
    [95000,70000]
]

LABELS = ["LOW","MEDIUM","HIGH","MEDIUM","HIGH"]

PLT.BOXPLOT(DATA, LABELS=LABELS)
PLT.TITLE("LOAN AMOUNT BY CREDIT SCORE")
```

```
PLT.YLABEL("LOAN AMOUNT")
PLT.SHOW()
```



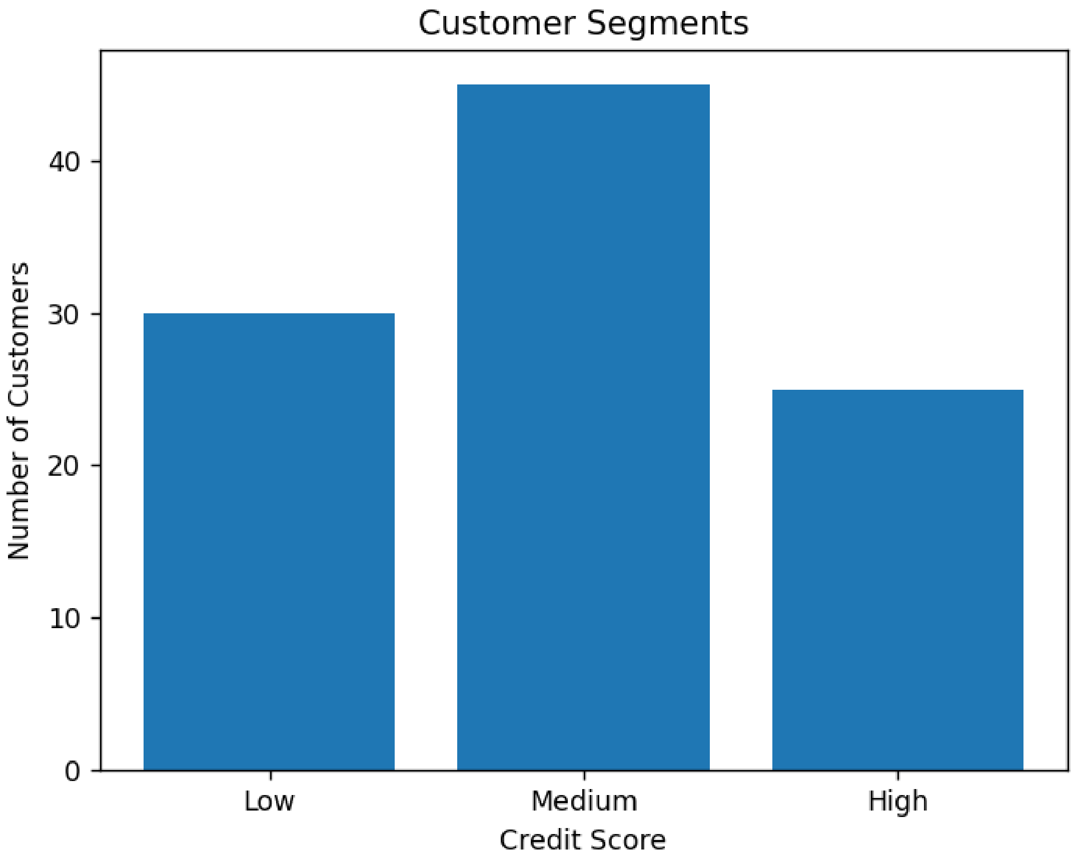
5.3. DENSITY PLOT

```
IMPORT SEABORN AS SNS
IMPORT MATPLOTLIB.PYPLOT AS PLT

LOAN = [50000,70000,60000,80000,90000,75000,65000,85000,95000,70000]

SNS.KDEPLOT(LOAN, FILL=TRUE)
PLT.TITLE("LOAN AMOUNT DISTRIBUTION")
PLT.XLABEL("LOAN AMOUNT")
PLT.SHOW()
```



5.4. BAR CHART

```
IMPORT MATPLOTLIB.PYPLOT AS PLT

CREDIT = ["LOW","MEDIUM","HIGH"]
CUSTOMERS = [30,45,25]

PLT.BAR(CREDIT, CUSTOMERS)
PLT.TITLE("CUSTOMER SEGMENTS")
PLT.XLABEL("CREDIT SCORE")
PLT.YLABEL("NUMBER OF CUSTOMERS")
PLT.SHOW()
```



Customer Segments
Number of Customers
0
10
20
30
40
Low
Medium
High
Credit Score